Learning Low-Rank Spectral Gradient Subspace With Structure Tensor Model for Guaranteed Hyperspectral Image Destriping and Denoising

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***Abstract*—In this article, based on tensor subspace modeling, we propose a guaranteed hyperspectral image (HSI) destriping and denoising method by learning a low-rank (LR) spectral gradient subspace with a structure tensor model (LLRSGSSTM) that aims to acquire the clean HSI from the noisy HSI with hybrid noise (containing Gaussian noise, impulse noise, and deadlines) and stripe noise. On one hand, by leveraging the inherent directionality of stripe noise, we aim to explore the physical property of the multimode gradients of stripe noise and propose the zero tensor property of the mode-1 gradient tensor of stripe noise as well as the low-tubal-rank attribute of mode-2 and mode-3 gradient tensors of stripe noise. On the other hand, by leveraging the spectral LR property of spectral gradient of HSI, we aim at learning the LR spectral gradient subspace (SGS) of HSI with structure tensor model, which decomposes the spectral gradient of HSI into the SGS representative coefficient (SGSRC) and the learnable SGS orthogonal basis matrix (SGSOBM). Moreover, the SGSRC indeed contains such important yet distinct spatial structures as corners, edges, and smooth regions, we novelly leverage the structure tensor to model these desired spatial structures of SGSRC and propose the unified SGSRC-based structure tensor total-variation (TV) prior of HSI. Preferably, the SGSRC with much smaller size can lead to much lower computational complexity than the HSI or the spectral gradient of HSI. Subsequently, we leverage the ADMM algorithm to optimize the LLRSGSSTM model and theoretically prove the convergence of LLRSGSSTM algorithm. Finally, extensive experiments are shown to verify the superiority of LLRSGSSTM for HSI destriping and denoising.**

***Index Terms*—Hyperspectral image (HSI) destriping and denoising, low-rank (LR), spectral gradient subspace representative coefficient (SGSRC), structure tensor, tensor low-tubal-rank prior.**

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NOMENCLATURE

z, $\mathbf{z}$, $\mathbf{Z}$, $\mathcal{Z}$	Scalar, vector, matrix, tensor.
$\mathcal{Z} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$	Order-3 tensor.
$\mathcal{Z}(i, j, k)$	(i, j, k)th element of $\mathcal{Z}$.
$\mathcal{Z}(i, j, :)$	(i, j)th tube of $\mathcal{Z}$.
$\mathcal{Z}(:, :, k)$ or $\mathbf{Z}_k$	kth mode-3 slice of $\mathcal{Z}$.
$\mathbf{Z}_{(1)} = \text{unfold}_1(\mathcal{Z}) \in \mathbb{R}^{n_1 \times n_2 n_3}$	Mode-1 unfolding of $\mathcal{Z}$.
$\mathbf{Z}_{(2)} = \text{unfold}_2(\mathcal{Z}) \in \mathbb{R}^{n_2 \times n_3 n_1}$	Mode-2 unfolding of $\mathcal{Z}$.
$\mathbf{Z}_{(3)} = \text{unfold}_3(\mathcal{Z}) \in \mathbb{R}^{n_3 \times n_1 n_2}$	Mode-3 unfolding of $\mathcal{Z}$.
$\mathcal{Z} = \text{fold}_1(\mathbf{Z}_{(1)}) \in \mathbb{R}^{n_1 \times n_2 \times n_3}$	Mode-1 folding of $\mathbf{Z}_{(1)}$.
$\mathcal{Z} = \text{fold}_2(\mathbf{Z}_{(2)}) \in \mathbb{R}^{n_1 \times n_2 \times n_3}$	Mode-2 folding of $\mathbf{Z}_{(2)}$.
$\mathcal{Z} = \text{fold}_3(\mathbf{Z}_{(3)}) \in \mathbb{R}^{n_1 \times n_2 \times n_3}$	Mode-3 folding of $\mathbf{Z}_{(3)}$.
$\overline{\mathcal{Z}} = \text{fft}(\mathcal{Z}, [\,], 3)$	Discrete Fourier transform (DFT) Along tubes of $\mathcal{Z}$.
$\mathcal{Z} = \text{ifft}(\overline{\mathcal{Z}}, [\,], 3)$	Inverse DFT along tubes of $\overline{\mathcal{Z}}$.
$\mathcal{A} = \mathcal{Z} \times_1 \mathbf{C} \Longleftrightarrow \mathbf{A}_{(1)} = \mathbf{C}\mathbf{Z}_{(1)}$	Mode-1 tensor-matrix product.
$\mathcal{A} = \mathcal{Z} \times_2 \mathbf{C} \Longleftrightarrow \mathbf{A}_{(2)} = \mathbf{C}\mathbf{Z}_{(2)}$	Mode-2 tensor-matrix product.
$\mathcal{A} = \mathcal{Z} \times_3 \mathbf{C} \Longleftrightarrow \mathbf{A}_{(3)} = \mathbf{C}\mathbf{Z}_{(3)}$	Mode-3 tensor-matrix product.
$\lvert \cdot \rvert$	Absolute value operator.
$\lVert \cdot \rVert_*$	Nuclear norm operator of the matrix
$\lVert \mathcal{Z} \rVert_1 = \sum_{i,j,k} \lvert \mathcal{Z}(i, j, k) \rvert$	l_1 norm of $\mathcal{Z}$.
$\lVert \mathcal{Z} \rVert_F = \sqrt{\sum_{i,j,k} \lvert \mathcal{Z}(i, j, k) \rvert^2}$	Frobenius norm of $\mathcal{Z}$.
$\lVert \mathcal{Z} \rVert_{\text{TNN}} = \sum_k \lVert \overline{\mathcal{Z}}(:, :, k) \rVert_*$	Tensor nuclear norm (TNN) of $\mathcal{Z}$.
$(\,\cdot\,)^T$	Transpose operator.

I. INTRODUCTION

BY CAPTURING the electromagnetic spectrum across hundreds of narrow bands, hyperspectral images (HSIs) enable precise material identification, driving advancements in the fields of precision agriculture and environmental monitoring. However, physical limitations during the acquisition process, including sensor malfunctions and atmospheric interference, inevitably introduce complex degradations. Specifically, the observed HSIs frequently suffer from a complicated

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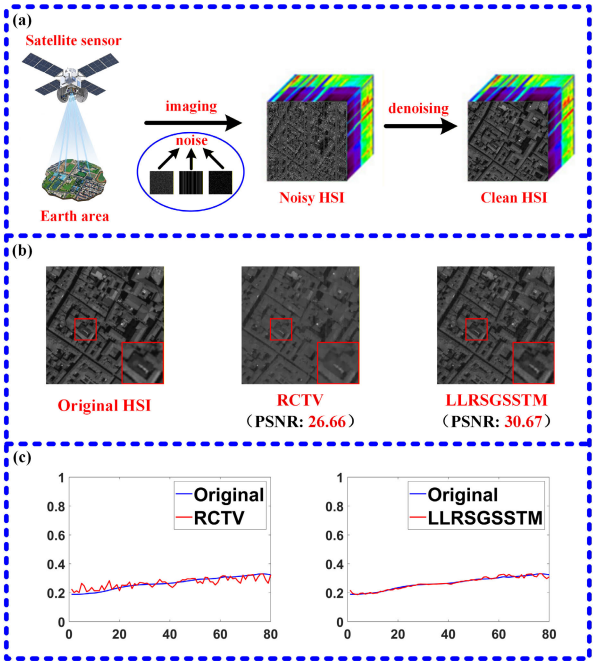


Fig. 1. (a) Flowchart of HSI denoising. (b) From left to right: original HSI, the restored HSI of RCTV [49], and the restored HSI of LLRSGSSTM. (c) From left to right: spectral curve fitting results of RCTV [49] and LLRSGSSTM.

mixture of stripe noise, Gaussian noise, impulse noise, and deadlines, which severely compromises spatial–spectral fidelity and degrades such downstream applications as classification [1], [2], [3], [4], [5] and target detection [6]. Consequently, HSI denoising (see Fig. 1) is an indispensable preprocessing step to recover high-fidelity HSI data for practical applications.

To mitigate the severe impacts of mixed noise and restore clean HSIs, extensive research efforts have been dedicated to developing effective denoising algorithms over the past decades. Broadly speaking, the mainstream methodologies can be categorized into two primary paradigms: model optimization-based approaches and deep-learning-based approaches. While these two paradigms rely on fundamentally distinct mechanisms to model the spatial–spectral dependencies of HSIs and characterize complex noise distributions, they share the same goal of restoring clean HSIs while preserving the intrinsic structural and spectral integrity of HSIs.

A. Deep-Learning-Based Approaches

In remote sensing, deep-learning-based methods have been successfully investigated [1], [2], [3], [4], [5], [6], [7], [8]. Specifically, deep-learning-based methods have recently appeared as a major paradigm in HSI denoising, leveraging the powerful representation capabilities of deep neural networks. By employing sophisticated architectures such as convolutional neural networks (CNNs), transformers, Mamba, and diffusion models [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], these deep-learning-based methods are designed to automatically learn complex and nonlinear mappings from degraded observations directly to their clean counterparts.

Despite demonstrating remarkable quantitative performance and high computational efficiency during the inference stage, the practical applicability of deep-learning-based approaches inherently remains constrained. Specifically, a primary bottleneck is their heavy reliance on large-scale paired training data, which significantly restricts their robust generalization capabilities. Consequently, these networks often encounter substantial challenges when dealing with the complex mixed noise distributions of real-world remote sensing scenarios.

B. Model Optimization-Based Approaches

Model optimization-based approaches formulate HSI denoising as an interpretable inverse problem and restore the clean HSI data by embedding intrinsic physical characteristics of HSI and different noise as regularization into the objective function. Within this paradigm, we specifically focus on regularization-based denoising models driven by four prominent priors, i.e., the local smoothness (LS) prior, the low-rank (LR) prior, the LS + LR prior, and the spectral subspace-based prior.

1) LS Prior: The fundamental principle of the LS prior stems from the continuous transitions typically observed across adjacent spatial pixels and contiguous spectral bands in HSIs. To mathematically model this piecewise smooth nature, regularization techniques applied to the image gradients are widely adopted. Among these, TV serves as a core prior that effectively mitigates local noise perturbations without blurring sharp edges. Consequently, numerous TV variants have been proposed to fully exploit the multidimensional smoothness of HSIs. Early representative methods include spectral–spatial adaptive TV [19] and spatiospectral TV [20]. To further improve edge preservation and structural fidelity during the denoising process, more complex gradient constraints have been progressively introduced, such as hybrid spatiospectral TV [21] and graph spatiospectral TV [22]. Specifically, these regularization mechanisms provide a robust restoration for maintaining the essential spatial–spectral continuity in the HSIs.

2) LR Prior: LR priors effectively capture the inherent spatial–spectral redundancies in HSIs. To formulate these structural correlations, existing LR-based denoising techniques can be broadly categorized into two paradigms: earlier matrix-based LR approaches that unfold the 3-D data into the 2-D matrix, and recent tensor-based LR approaches that directly preserve the multiway tensor nature of HSIs.

Matrix-Based LR Methods: To leverage matrix-based LR priors, 3-D HSIs are conventionally unfolded into 2-D matrices via spectral unfolding or nonlocal patch grouping. Building upon this 2-D representation, representative methods formulate the denoising task as an LR matrix recovery or double decomposition problem [23], [24]. To address the complex degradations encountered in real-world scenarios, sophisticated statistical strategies have been introduced to robustly model mixture, non-i.i.d., and asymmetric noise distributions [25], [26], [27]. Furthermore, to alleviate the suboptimal estimation bias inherent in traditional LR-based nuclear norm minimization, nonconvex LR approximations are preferably adopted, providing a more accurate constraint on the essential

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spatial–spectral structures of HSIs [28], [29]. Moreover, LR-based graph Laplacian regularization has been introduced to efficiently capture the intrinsic manifold structures of the unfolded matrices [30], [31].

Tensor-Based LR Methods: While matrix-based methods are effective, the unfolding operations inevitably disrupt the inherent 3-D spatial–spectral geometry of HSIs. To preserve these multidimensional correlations, tensor-based LR methods directly model the HSI as an order-3 tensor. Among these approaches, several frameworks employ LR tensor recovery and factorization techniques, incorporating regularizations like double-factor and low-fibered-rank constraints to restore clean HSIs from mixed noise [32], [33], [34]. To further exploit internal image redundancies, combining nonlocal self-similarity priors with tensor decomposition or structured sparsity has proven high effectiveness in preserving spatial textures [35], [36]. Moreover, the tensor ring decomposition is widely utilized to capture multidirectional correlations [37], [38].

3) LS + LR Prior: To further enhance the denoising performance of HSIs, researchers have extensively explored the joint modeling of LS and LR priors. The majority of existing methods adopt a compound regularization strategy by combining TV and LR matrix or tensor constraints as independent regularization terms. Representative works include TV-regularized LR matrix factorization (LRTV) [39], along with various extended frameworks incorporating tensor decomposition [40], nonlocal priors [41], [42], global variation [43], factor group sparsity [44], and anisotropic spatial–spectral TV regularized double LR approximation [45]. Furthermore, to capture both LS and LR priors simultaneously within a single framework, unified regularization terms have been proposed, such as 3-D correlated TV (3DCTV) [46] and tensor correlated TV (t-CTV) [47], which achieve more efficient joint structural modeling in the matrix and tensor domains, respectively.

4) Spectral Subspace-Based Prior: The spectral subspace-based prior provides a computationally efficient denoising paradigm by projecting the HSI in the original or gradient domain into a low-dimensional subspace and optimizing the corresponding representative coefficient. In the original domain, fundamental frameworks typically impose constraints such as low-rankness, sparsity, and TV on the representative coefficient to restore the clean HSI [48], [49], [50], [51]. In gradient domain, the enhanced 3-D TV (E-3DTV) [52] approach explicitly models the gradient sparsity of the representative coefficient across spatial and spectral dimensions, and the hyper-Laplacian TV in the spectral gradient domain (HLTVSG) [53] introduces a hyper-Laplacian TV of the representative coefficient in the spectral gradient subspace (SGS), which more accurately fits the heavy-tailed distribution of spatial gradients of the representative coefficient, thereby robustly enforcing LS while preserving sharp edges.

C. Our Contributions

1) Motivations: In this article, owing to the inherent directionality of stripe noise, we model the stripe noise independently and mainly investigate HSI destriping and denoising simultaneously. However, on one hand, most of the above approaches do not model the stripe noise independently; they still model the stripe noise as sparse noise. On the other hand, although several approaches model the stripe noise independently, they impose the common and simple band-wise LR matrix prior of stripe noise [24], [30], [31], [45], [53]. Indeed, the stripe noise is an order-3 tensor and holds strongly structural and directional properties. Thus, in this article, we particularly view the stripe noise as an order-3 tensor, and explore the tensor-based structural and directional properties of the stripe noise simultaneously.

Moreover, although the spectral subspace prior-based approaches show much higher computational efficiency. However, in the spectral subspace, RCTV imposes the spatial TV on the representative coefficient of the HSI [49], E-3DTV imposes the spatial gradient sparsity on the representative coefficient of the spatial and spectral gradients of HSI [52], and HLTVSG imposes the spatial hyper-Laplacian TV on the representative coefficient of the spectral gradient of HSI [53], where the individual representative coefficient is indeed an image and even an order-3 tensor, which also has distinct important structures, such as corners, edges, and smooth regions. However, the structural prior modeling for these distinct important structures (i.e., corners, edges, and smooth regions) of the representative coefficient is not deeply explored in RCTV, E-3DTV, and HLTVSG. Thus, in this article, we particularly investigate the spectral subspace prior-based approach and aim to explore the structural prior modeling for these distinct important structures (i.e., corners, edges, and smooth regions) of the representative coefficient in the spectral subspace.

2) Contributions: To address these difficulties, in this article, based on tensor subspace modeling, we propose a guaranteed HSI destriping and denoising method by learning LR SGS with structure tensor model (LLRSGSSTM), whose modeling flowchart is displayed in Fig. 2.

On one hand, by leveraging the inherent directionality of stripe noise, we aim to explore the physical property of the multimode gradient tensors of stripe noise, and propose the zero tensor property of the mode-1 gradient tensor of stripe noise as well as the low-tubal-rank property of the mode-2 and mode-3 gradient tensors of stripe noise, which is unified as a multimode gradient tensor prior term of stripe noise.

On the other hand, by leveraging the spectral LR attribute of spectral gradient of HSI, we aim at learning the LR SGS of HSI with structure tensor model, which decomposes the spectral gradient of HSI into the SGS representative coefficient (SGSRC) and the learnable SGS orthogonal basis matrix (SGSOBM). Moreover, the SGSRC indeed contains such distinct yet important spatial structures as corners, edges, and smooth regions, we novelly leverage the structure tensor to model these desired spatial structures of SGSRC and impose the structure tensor TV (STV) prior on the SGSRC, and propose the unified SGSRC-based STV prior term of HSI. Preferably, the SGSRC with much smaller size can lead to much lower computational complexity than the HSI or the spectral gradient of HSI.

Subsequently, we leverage the alternating direction method of multiplier (ADMM) algorithm to optimize the LLRS-

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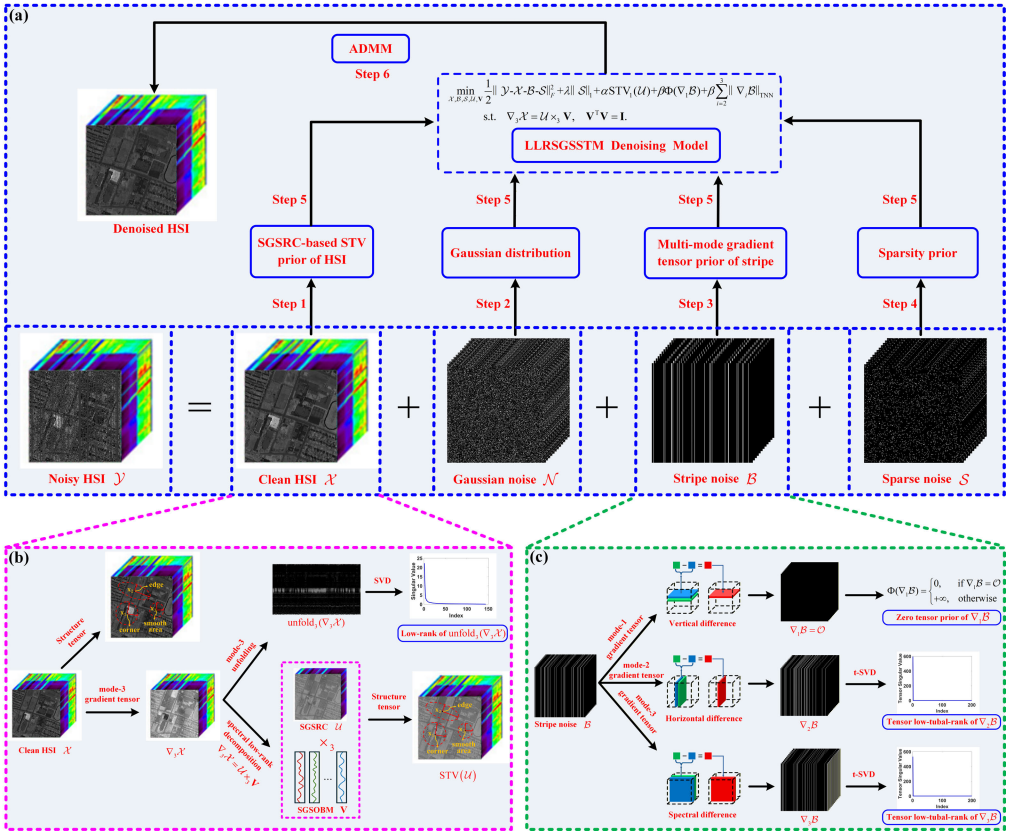


Fig. 2. Modeling flowsheets of (a) proposed LLRSGSSTM method, (b) SGSRC-based STV prior of HSI $\mathcal{X}$, and (c) multimode gradient tensor prior of stripe noise $\mathcal{B}$.

GSSTM model and theoretically prove the convergence of LLRSGSSTM algorithm.

Finally, we show extensive experiments to verify the superiority of LLRSGSSTM for HSI destriping and denoising.

The article is organized as follows. Sections II and III illustrate the modeling and optimization procedures of LLRSGSSTM, respectively. Section IV illustrates the experiments. Section V provides the conclusion.

II. Proposed LLRSGSSTM Model

A. Notations and Definition

We define some notations in nomenclature.

Definition 1 (Gradient Tensors [47]): For an order-3 tensor $\mathcal{Z} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, then the mode-$i$ gradient tensors of $\mathcal{Z}$, i.e., $\nabla_i \mathcal{Z} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, $i = 1, 2, 3$, are defined as

$$\nabla_i \mathcal{Z} = \mathcal{Z} \times_i \mathbf{D}_{n_i}, \quad i = 1, 2, 3 \tag{1}$$

where ∇_i is the gradient operator along the ith mode. $\mathcal{Z} \times_i \mathbf{D}_{n_i}$ is the mode-i tensor–matrix product of $\mathcal{Z}$ with $\mathbf{D}_{n_i}$. $\mathbf{D}_{n_i} \in \mathbb{R}^{n_i \times n_i}$ is the row circulant matrix of $(-1, 1, 0, \ldots, 0) \in \mathbb{R}^{1 \times n_i}$, i.e.,

$$\mathbf{D}_{n_i} = \begin{bmatrix} -1 & 1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & -1 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & -1 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & -1 & 1 \\ 1 & 0 & 0 & 0 & \cdots & 0 & -1 \end{bmatrix}_{n_i \times n_i}. \tag{2}$$

Definition 2 (t-CTV [47]): For an order-3 tensor $\mathcal{Z} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, the t-CTV of $\mathcal{Z}$ is defined as

$$\text{t-CTV}(\mathcal{Z}) = \sum_{i=1}^{3} \|\nabla_i \mathcal{Z}\|_{\text{TNN}}. \tag{3}$$

B. Hybrid Noise Model Formulation

Based on the unified tensor form, the HSI with hybrid noise pollution, including stripe noise, sparse noise (i.e., impulse noise and deadlines), and Gaussian noise, is modeled as

$$\mathcal{Y} = \mathcal{X} + \mathcal{B} + \mathcal{S} + \mathcal{N} \tag{4}$$

where $\mathcal{Y} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ denotes the noisy HSI, $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ denotes the clean HSI to be restored, $\mathcal{B} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ denotes the stripe noise, $\mathcal{S} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ denotes the sparse noise, and $\mathcal{N} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ denotes the Gaussian noise.

C. Multimode Gradient Tensor Prior of Stripe Noise

Indeed, the stripe noise $\mathcal{B} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is an order-3 tensor and holds strongly structural and directional properties. Different from the common and simple band-wise LR matrix prior of stripe noise [24], [30], [31], [45], [53], i.e., $\sum_{i_3=1}^{n_3} \|\mathcal{B}(:, :, i_3)\|_*$, we directly view the stripe noise $\mathcal{B} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ as an order-3 tensor in this article, and explore the tensor-based structural and directional properties of the stripe noise $\mathcal{B}$ simultaneously.

To this end, according to the vertical property of stripe noise $\mathcal{B}$, we explore the tensor-based structural and directional properties of the stripe noise $\mathcal{B}$ simultaneously in the multimode

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gradient domains, i.e., $\nabla_1\mathcal{B} \in \mathbb{R}^{n_1\times n_2\times n_3}$, $\nabla_2\mathcal{B} \in \mathbb{R}^{n_1\times n_2\times n_3}$, and $\nabla_3\mathcal{B} \in \mathbb{R}^{n_1\times n_2\times n_3}$ (see Definition 1).

1) As shown in Fig. 2, the mode-1 gradient $\nabla_1\mathcal{B} \in \mathbb{R}^{n_1\times n_2\times n_3}$ has the zero tensor constraint, i.e.,

$$\nabla_1\mathcal{B} = \mathcal{O} \tag{5}$$

where $\mathcal{O} \in \mathbb{R}^{n_1\times n_2\times n_3}$ is the zero tensor.
Then, we impose this zero tensor constraint $\nabla_1\mathcal{B} = \mathcal{O}$ as an indicator function $\Phi(\nabla_1\mathcal{B})$, i.e.,

$$\Phi(\nabla_1\mathcal{B}) = \begin{cases} 0, & \text{if } \nabla_1\mathcal{B} = \mathcal{O} \\ +\infty, & \text{otherwise.} \end{cases} \tag{6}$$

2) As shown in Fig. 2, for the mode-2 gradient $\nabla_2\mathcal{B} \in \mathbb{R}^{n_1\times n_2\times n_3}$ as well as the mode-3 gradient $\nabla_3\mathcal{B} \in \mathbb{R}^{n_1\times n_2\times n_3}$, we aim to explore their underlying tensor-based low-rank properties by employing the tensor singular value decomposition (t-SVD) [33] of $\nabla_2\mathcal{B}$ and $\nabla_3\mathcal{B}$, and obtain the tensor low-tubal-rank properties of $\nabla_2\mathcal{B}$ and $\nabla_3\mathcal{B}$, i.e., the tensor tubal rank of $\nabla_2\mathcal{B}$ and $\nabla_3\mathcal{B}$ denoted as $\text{rank}_\text{t}(\nabla_2\mathcal{B})$ and $\text{rank}_\text{t}(\nabla_3\mathcal{B})$ are low, which are defined as [33]

$$\begin{aligned} \text{rank}_\text{t}(\nabla_2\mathcal{B}) &= \sharp\{i, \mathcal{D}_2(i,i,:) \neq \mathbf{0} \in \mathbb{R}^{n_3}\} \\ \text{rank}_\text{t}(\nabla_3\mathcal{B}) &= \sharp\{i, \mathcal{D}_3(i,i,:) \neq \mathbf{0} \in \mathbb{R}^{n_3}\} \end{aligned} \tag{7}$$

where $\nabla_2\mathcal{B} = \mathcal{C}_2 * \mathcal{D}_2 * \mathcal{E}_2^T$ and $\nabla_3\mathcal{B} = \mathcal{C}_3 * \mathcal{D}_3 * \mathcal{E}_3^T$ are, respectively, the t-SVD of $\nabla_2\mathcal{B}$ and $\nabla_3\mathcal{B}$; $\sharp\{\cdot\}$ is the cardinality operator; and $\mathbf{0} \in \mathbb{R}^{n_3}$ is the zero vector.

Then, by utilizing the tensor nuclear norm (TNN) [33] for convex approximation of $\text{rank}_\text{t}(\nabla_2\mathcal{B})$ and $\text{rank}_\text{t}(\nabla_3\mathcal{B})$, we construct the TNN-based tensor low-tubal-rank prior term of $\nabla_2\mathcal{B}$ and $\nabla_3\mathcal{B}$ as

$$\mathcal{R}_1(\mathcal{B}) = \beta\sum_{i=2}^{3}\|\nabla_i\mathcal{B}\|_\text{TNN}. \tag{8}$$

Therefore, we construct the unified multimode gradient tensor prior term $\mathcal{B}$ of stripe noise as

$$\mathcal{R}(\mathcal{B}) = \beta\Phi(\nabla_1\mathcal{B}) + \beta\sum_{i=2}^{3}\|\nabla_i\mathcal{B}\|_\text{TNN}. \tag{9}$$

Remark 1: Since $\nabla_1\mathcal{B} = \mathcal{O}$, then we can obtain that $\|\nabla_1\mathcal{B}\|_\text{TNN} = 0$. In this sense, the unified multimode gradient tensor prior term of stripe noise $\mathcal{B}$ proposed in (9) can be alternatively rewritten as $\mathcal{R}(\mathcal{B}) = \beta\sum_{i=1}^{3}\|\nabla_i\mathcal{B}\|_\text{TNN} = \beta\,\text{t-CTV}(\mathcal{B})$, i.e., the t-CTV of $\mathcal{B}$ (see Definition 2).

D. SGSRC-Based STV Prior of HSI

1) Motivation From RCTV of HSI: Specifically, the HSI $\mathcal{X} \in \mathbb{R}^{n_1\times n_2\times n_3}$ has the spectral LR property, i.e., the mode-3 unfolded HSI $\text{unfold}_3(\mathcal{X}) \in \mathbb{R}^{n_3\times n_1n_2}$ is LR. Based on the spectral LR factorization, the representative coefficient TV (RCTV) [49] method factorizes $\mathcal{X}$ into the spectral subspace as

$$\mathcal{X} = \widetilde{\mathcal{U}} \times_3 \widetilde{\mathbf{V}} \tag{10}$$

where $\widetilde{\mathcal{U}} \in \mathbb{R}^{n_1\times n_2\times r}$ is the representative coefficient (RC). $\widetilde{\mathbf{V}} \in \mathbb{R}^{n_3\times r}$ is the column orthogonal spectral basis matrix (COSBM) satisfying $\widetilde{\mathbf{V}}^T\widetilde{\mathbf{V}} = \mathbf{I} \in \mathbb{R}^{r\times r}$ and $\text{rank}(\widetilde{\mathbf{V}}) = \text{rank}(\text{unfold}_3(\mathcal{X})) = r$, $r \ll n_3$.

Thus, according to $\widetilde{\mathbf{V}}^T\widetilde{\mathbf{V}} = \mathbf{I} \in \mathbb{R}^{r\times r}$, the RC $\widetilde{\mathcal{U}}$ can be derived as $\widetilde{\mathcal{U}} = \mathcal{X} \times_3 \widetilde{\mathbf{V}}^T$. Furthermore, based on the LS prior of RC $\widetilde{\mathcal{U}} \in \mathbb{R}^{n_1\times n_2\times r}$ by imposing the spatial TV, the RCTV of HSI $\mathcal{X} \in \mathbb{R}^{n_1\times n_2\times n_3}$ is defined as [49]

$$\begin{aligned} \|\mathcal{X}\|_\text{RCTV} &= \sum_{i=1}^{r}\left\|\widetilde{\mathcal{U}}(:,:,i)\right\|_\text{TV} \\ &= \sum_{i=1}^{r}\sum_{j=1}^{2}\left\|\nabla_j\widetilde{\mathcal{U}}(:,:,i)\right\|_1. \end{aligned} \tag{11}$$

Remark 2: By comparing with the prior on HSI $\mathcal{X} \in \mathbb{R}^{n_1\times n_2\times n_3}$ in the original domain, RCTV imposes the LS prior of RC $\widetilde{\mathcal{U}} \in \mathbb{R}^{n_1\times n_2\times r}$ in the subspace, it can perform good denoising performance and particularly show higher computational efficiency with much lower computational complexity of $O(n_1n_2r)$.

Nevertheless, RCTV only imposes the spatial TV prior on each mode-3 slice of RC $\widetilde{\mathcal{U}} \in \mathbb{R}^{n_1\times n_2\times r}$, which can well preserve edges. However, for each mode-3 slice of RC $\widetilde{\mathcal{U}}$, it indeed has different structures such as edges, corners, and smooth regions, which should be simultaneously considered for modeling. To this end, we aim to explore more suitable prior models for these important structures of RC in the subspace for HSIs.

2) Proposed SGSRC-Based STV Prior of HSI: Based on the spectral low-rankness of $\mathcal{X} \in \mathbb{R}^{n_1\times n_2\times n_3}$ in the original domain, this article focuses on the spectral gradient of HSI denoted as $\nabla_3\mathcal{X} \in \mathbb{R}^{n_1\times n_2\times n_3}$, i.e., mode-3 gradient tensor $\nabla_3\mathcal{X} = \mathcal{X} \times_3 \mathbf{D}_{n_3}$, where $\mathbf{D}_{n_3} \in \mathbb{R}^{n_3\times n_3}$ is defined in (2). Since the mode-3 gradient operator ∇_3 is linear and $\mathbf{D}_{n_3} \in \mathbb{R}^{n_3\times n_3}$ is full-rank, the mode-3 gradient tensor $\nabla_3\mathcal{X}$ inherits the spectral low-rankness, i.e., $\text{unfold}_3(\nabla_3\mathcal{X}) \in \mathbb{R}^{n_3\times n_1n_2}$ is LR (see Fig. 2), where $\text{rank}(\text{unfold}_3(\nabla_3\mathcal{X})) = \text{rank}(\text{unfold}_3(\mathcal{X})) = r \ll n_3$.

Thus, by applying the spectral LR factorization of $\nabla_3\mathcal{X}$, we can obtain the SGS representation as

$$\nabla_3\mathcal{X} = \mathcal{U} \times_3 \mathbf{V} \tag{12}$$

where $\mathcal{U} \in \mathbb{R}^{n_1\times n_2\times r}$ denotes the SGSRC. $\mathbf{V} \in \mathbb{R}^{n_3\times r}$ denotes the SGSOBM satisfying $\mathbf{V}^T\mathbf{V} = \mathbf{I} \in \mathbb{R}^{r\times r}$ and $\text{rank}(\mathbf{V}) = r \ll n_3$.

As we can evidently observe from Fig. 2, in the original domain, the HSI $\mathcal{X} \in \mathbb{R}^{n_1\times n_2\times n_3}$ has distinct important structures, such as corners, edges, and smooth regions. Moreover, in the SGS, the SGSRC $\mathcal{U} \in \mathbb{R}^{n_1\times n_2\times r}$ also has distinct important structures, such as corners, edges, and smooth regions (see Fig. 2). Thanks to the representation capacity of structure tensor-based image descriptor [54] for suitably modeling corners, edges, and smooth regions, lots of structure tensor-based methods are successfully applied for image structure modeling [55], [56], [57]. Therefore, this article leverages the structure tensor for modeling these distinct important corners, edges, and smooth regions of the SGSRC $\mathcal{U}$, and propose the SGSRC-based STV prior term.

Indeed, the structure tensor of SGSRC $\mathcal{U} \in \mathbb{R}^{n_1\times n_2\times r}$ at a spatial pixel $\mathbf{x}_n \in \mathbb{R}^2$, $n = 1, 2, \ldots, n_1n_2$, i.e., $S_\mathbf{H}\mathcal{U}(\mathbf{x}_n) \in \mathbb{R}^{2\times 2}$, is defined as

$$S_\mathbf{H}\mathcal{U}(\mathbf{x}_n) = \mathbf{H} * \left[\mathcal{J}\mathcal{U}^T\mathcal{J}\mathcal{U}\right](\mathbf{x}_n) \tag{13}$$

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where the Gaussian kernel $\mathbf{H} \in \mathbb{R}^{d\times d}$ has the spatial support $[-d_{\mathbf{H}}, d_{\mathbf{H}}]\times[-d_{\mathbf{H}}, d_{\mathbf{H}}]$ so that $d = 2d_{\mathbf{H}}+1$. $*$ is the convolution operator. The Jacobian matrix $\mathcal{JU}(\mathbf{x}_n) \in \mathbb{R}^{r\times 2}$ is defined as

$$\mathcal{JU}(\mathbf{x}_n) = \left(\nabla\mathbf{U}_1(\mathbf{x}_n), \ldots, \nabla\mathbf{U}_r(\mathbf{x}_n)\right)^T \tag{14}$$

where $\nabla\mathbf{U}_k(\mathbf{x}_n) = (\nabla_1\mathbf{U}_k(\mathbf{x}_n), \nabla_2\mathbf{U}_k(\mathbf{x}_n))^T \in \mathbb{R}^2$, in which ∇_1 and ∇_2 are mode-1 and mode-2 gradient operators, $k = 1, 2, \ldots, r$.

Moreover, the structure tensor $S_{\mathbf{H}}\mathcal{U}(\mathbf{x}_n) \in \mathbb{R}^{2\times 2}$ is equivalently defined as [54]

$$S_{\mathbf{H}}\mathcal{U}(\mathbf{x}_n) = \left[\mathcal{J}_{\mathbf{H}}\mathcal{U}^T\mathcal{J}_{\mathbf{H}}\mathcal{U}\right](\mathbf{x}_n) \tag{15}$$

where the patch-based Jacobian matrix $\mathcal{J}_{\mathbf{H}}\mathcal{U}(\mathbf{x}_n) \in \mathbb{R}^{d^2r\times 2}$ [54] is defined as

$$\mathcal{J}_{\mathbf{H}}\mathcal{U}(\mathbf{x}_n) = \left(\widetilde{\nabla}\mathbf{U}_1(\mathbf{x}_n), \ldots, \widetilde{\nabla}\mathbf{U}_r(\mathbf{x}_n)\right)^T \tag{16}$$

where $\widetilde{\nabla}\mathbf{U}_k(\mathbf{x}_n) = (w_1\nabla\mathbf{U}_k(\mathbf{x}_n), \ldots, w_{d^2}\nabla\mathbf{U}_k(\mathbf{x}_n)) \in \mathbb{R}^{2\times d^2}$. $w_i = (\mathbf{H}(\mathbf{x}_n - \mathbf{y}_i))^{1/2}$, in which $\mathbf{y}_i$ is the ith neighbor of $\mathbf{x}_n$, $i = 1, 2, \ldots, d^2$.

For the symmetric and semi-positive structure tensor $S_{\mathbf{H}}\mathcal{U}(\mathbf{x}_n) \in \mathbb{R}^{2\times 2}$, its two nonnegative eigenvalues are computed as λ_n^+ and λ_n^-, where $\lambda_n^+ \geq \lambda_n^- \geq 0$. Then, the corners, edges, and smooth regions of SGSRC $\mathcal{U}$ at $\mathbf{x}_n$ are modeled by $S_{\mathbf{H}}\mathcal{U}(\mathbf{x}_n)$ via the eigenvalues λ_n^+ and λ_n^-.

1) When both λ_n^+ and λ_n^- are larger, $\mathbf{x}_n$ is modeled as the corner.
2) When λ_n^+ is larger and λ_n^- is smaller, $\mathbf{x}_n$ is modeled as the edge.
3) When both λ_n^+ and λ_n^- are smaller, $\mathbf{x}_n$ is modeled as the smooth region.

Based on (15), the two singular values of $\mathcal{J}_{\mathbf{H}}\mathcal{U}(\mathbf{x}_n) \in \mathbb{R}^{d^2r\times 2}$ are computed as $(\lambda_n^+)^{1/2}$ and $(\lambda_n^-)^{1/2}$. Therefore, we visualize the structure tensor of SGSRC $\mathcal{U}$ at each point $\mathbf{x}_n$ by utilizing an ellipse with major radius $(\lambda_n^+)^{1/2}$ and minor radius $(\lambda_n^-)^{1/2}$. Moreover, Fig. 2 illustrates the structure tensor of SGSRC $\mathcal{U}$ at $\mathbf{x}_1$ (corner), $\mathbf{x}_2$ (edge), and $\mathbf{x}_3$ (smooth region), and the flowchart of STV prior of SGSRC $\mathcal{U}$.

Consequently, relied on $(\lambda_n^+)^{1/2}$ and $(\lambda_n^-)^{1/2}$ of $\mathcal{J}_{\mathbf{H}}\mathcal{U}(\mathbf{x}_n) \in \mathbb{R}^{d^2r\times 2}$, the STV prior of SGSRC $\mathcal{U}$ is established as [54]

$$R(\mathcal{U}) = \mathrm{STV}_p(\mathcal{U}) = \sum_{n=1}^{n_1n_2}\|\mathcal{J}_{\mathbf{H}}\mathcal{U}(\mathbf{x}_n)\|_{S_p} \tag{17}$$

where the Schatten-p norm of the matrix $\mathbf{Z}$ is defined as $\|\mathbf{Z}\|_{S_p} = \left(\sum_j(\sigma_j(\mathbf{Z}))^p\right)^{1/p}$, $\sigma_j(\mathbf{Z})$ is the jth singular value of $\mathbf{Z}$, and $p \geq 1$.

Furthermore, in the unified spatial gradient subspace framework of $\nabla_3\mathcal{X} = \mathcal{U}\times_3\mathbf{V}$ and $\mathbf{V}^T\mathbf{V} = \mathbf{I}$, i.e., $\mathcal{U} = \nabla_3\mathcal{X}\times_3\mathbf{V}^T$, we establish the SGSRC-based STV prior of HSI $\mathcal{X}$ as

$$\mathcal{R}(\mathcal{X}) = \mathrm{STV}_1(\mathcal{U}) \quad \text{s.t. } \nabla_3\mathcal{X} = \mathcal{U}\times_3\mathbf{V},\ \mathbf{V}^T\mathbf{V} = \mathbf{I}. \tag{18}$$

Remark 3: Since the STV of $p > 1$ often performs worse than the STV of $p = 1$ [54], this article chooses $p = 1$ in the SGSRC-based STV prior of HSI $\mathcal{X}$ (18).

E. Proposed Model

Based on the unified multimode gradient tensor prior term of stripe noise $\mathcal{B}$ in (9) and the SGSRC-based STV prior of HSI $\mathcal{X}$ in (18), we aim at LLRSGSSTM for HSI denoising and destriping as

$$\begin{aligned}\left(\hat{\mathcal{X}}, \hat{\mathcal{B}}, \hat{\mathcal{S}}, \hat{\mathcal{U}}, \hat{\mathbf{V}}\right) = \arg\min_{\mathcal{X},\mathcal{B},\mathcal{S},\mathcal{U},\mathbf{V}} &\frac{1}{2}\|\mathcal{Y} - \mathcal{X} - \mathcal{B} - \mathcal{S}\|_F^2 + \lambda\|\mathcal{S}\|_1 \\ &+ \alpha\,\mathrm{STV}_1(\mathcal{U}) + \beta\Phi(\nabla_1\mathcal{B}) + \beta\sum_{i=2}^{3}\|\nabla_i\mathcal{B}\|_{\mathrm{TNN}} \\ &\text{s.t.} \quad \nabla_3\mathcal{X} = \mathcal{U}\times_3\mathbf{V}, \quad \mathbf{V}^T\mathbf{V} = \mathbf{I}\end{aligned} \tag{19}$$

where $1/2\|\mathcal{Y} - \mathcal{X} - \mathcal{B} - \mathcal{S}\|_F^2$ models the Gaussian noise $\mathcal{N}$. $\|\mathcal{S}\|_1$ models the sparse noise $\mathcal{S}$. λ, α, and β are the regularization parameters.

III. Optimization Algorithm

A. Model Optimization Procedure

By introducing auxiliary variables $\mathcal{A}$ and $\mathcal{G}_i$ ($i = 1, 2, 3$), we equivalently transform the LLRSGSSTM (19) into a constrained optimization problem, which enables the efficient application of the ADMM algorithm and is given as

$$\begin{aligned}\min_{\mathcal{X},\mathcal{B},\mathcal{S},\mathcal{A},\mathcal{G}_i,\mathcal{U},\mathbf{V}} &\frac{1}{2}\|\mathcal{Y} - \mathcal{X} - \mathcal{B} - \mathcal{S}\|_F^2 + \lambda\|\mathcal{S}\|_1 + \alpha\,\mathrm{STV}_1(\mathcal{A}) \\ &+ \beta\Phi(\mathcal{G}_1) + \beta\sum_{i=2}^{3}\|\mathcal{G}_i\|_{\mathrm{TNN}} \\ &\text{s.t.} \quad \nabla_3\mathcal{X} = \mathcal{U}\times_3\mathbf{V}, \quad \mathbf{V}^T\mathbf{V} = \mathbf{I} \\ &\quad \mathcal{U} = \mathcal{A}, \quad \nabla_i\mathcal{B} = \mathcal{G}_i, \quad i = 1, 2, 3.\end{aligned} \tag{20}$$

Consequently, the corresponding augmented Lagrangian function is constructed as

$$\begin{aligned}f(\mathcal{X}, \mathcal{B}, \mathcal{S}, \mathcal{A}, \mathcal{G}_i, \mathcal{U}, \mathbf{V}, \mathcal{D}, \mathcal{P}, \mathcal{H}_i) = &\frac{1}{2}\|\mathcal{Y} - \mathcal{X} - \mathcal{B} - \mathcal{S}\|_F^2 \\ &+ \lambda\|\mathcal{S}\|_1 + \alpha\,\mathrm{STV}_1(\mathcal{A}) + \beta\Phi(\mathcal{G}_1) + \beta\sum_{i=2}^{3}\|\mathcal{G}_i\|_{\mathrm{TNN}} \\ &+ \frac{\mu}{2}\|\nabla_3\mathcal{X} - \mathcal{U}\times_3\mathbf{V} + \mathcal{D}\|_F^2 + \frac{\mu}{2}\|\mathcal{U} - \mathcal{A} + \mathcal{P}\|_F^2 \\ &+ \frac{\mu}{2}\sum_{i=1}^{3}\|\nabla_i\mathcal{B} - \mathcal{G}_i + \mathcal{H}_i\|_F^2\end{aligned} \tag{21}$$

where the Lagrange multipliers are denoted by $\mathcal{D}$, $\mathcal{P}$, and $\mathcal{H}_i$ ($i = 1, 2, 3$). μ acts as the penalty parameter.

Relied on the ADMM scheme, the iterative optimization of model (20) can be decomposed into several independent minimization subproblems.

1) Update $\mathcal{S}$: The $\mathcal{S}^{(k+1)}$ subproblem is deduced from (21) as

$$\mathcal{S}^{(k+1)} = \arg\min_{\mathcal{S}}\frac{1}{2}\left\|\mathcal{Y} - \mathcal{X}^{(k)} - \mathcal{B}^{(k)} - \mathcal{S}\right\|_F^2 + \lambda\|\mathcal{S}\|_1 \tag{22}$$

which is optimized by leveraging the soft-thresholding operator as

$$\mathcal{S}^{(k+1)} = \mathrm{shrink}\left(\mathcal{Y} - \mathcal{X}^{(k)} - \mathcal{B}^{(k)}, \lambda\right) \tag{23}$$

where $\mathrm{shrink}(\mathcal{J}, \lambda) = \mathrm{sign}(\mathcal{J})\max(0, |\mathcal{J}| - \lambda)$.

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2) Update $\mathcal{A}$: The $\mathcal{A}^{(k+1)}$ subproblem is deduced from (21) as

$$\mathcal{A}^{(k+1)} = \arg\min_{\mathcal{A}} \alpha \,\mathrm{STV}_1(\mathcal{A}) + \frac{\mu}{2}\left\|\mathcal{U}^{(k)} - \mathcal{A} + \mathcal{P}^{(k)}\right\|_F^2. \quad (24)$$

By leveraging the STV-based proximal algorithm [54], $\mathcal{A}^{(k+1)}$ can be obtained as

$$\mathcal{A}^{(k+1)} = \mathrm{prox}_{\mathrm{STV}}\left(\mathcal{U}^{(k)} + \mathcal{P}^{(k)}, \alpha/\mu\right). \quad (25)$$

3) Update $\mathcal{U}$: The $\mathcal{U}^{(k+1)}$ subproblem is deduced from (21) as

$$\mathcal{U}^{(k+1)} = \arg\min_{\mathcal{U}} \frac{\mu}{2}\left\|\nabla_3\mathcal{X}^{(k)} - \mathcal{U}\times_3\mathbf{V}^{(k)} + \mathcal{D}^{(k)}\right\|_F^2 + \frac{\mu}{2}\left\|\mathcal{U} - \mathcal{A}^{(k+1)} + \mathcal{P}^{(k)}\right\|_F^2 \quad (26)$$

which admits a closed-form solution as

$$\mathcal{U}^{(k+1)} = \frac{\left(\nabla_3\mathcal{X}^{(k)} + \mathcal{D}^{(k)}\right)\times_3\mathbf{V}^{(k)^T} + \mathcal{A}^{(k+1)} - \mathcal{P}^{(k)}}{2}. \quad (27)$$

4) Update $\boldsymbol{V}$: The $\mathbf{V}^{(k+1)}$ subproblem is deduced from (21) as

$$\mathbf{V}^{(k+1)} = \arg\min_{\mathbf{V},\mathbf{V}^T\mathbf{V}=\mathbf{I}} \frac{\mu}{2}\left\|\nabla_3\mathcal{X}^{(k)} - \mathcal{U}^{(k+1)}\times_3\mathbf{V} + \mathcal{D}^{(k)}\right\|_F^2. \quad (28)$$

By leveraging the condensed singular value decomposition (SVD), the solution of (28) can be formulated as [58]

$$\begin{cases}\mathbf{W}^{(k)} = \mathrm{unfold}_3\left(\nabla_3\mathcal{X}^{(k)} + \mathcal{D}^{(k)}\right)\left(\mathrm{unfold}_3\left(\mathcal{U}^{(k+1)}\right)\right)^T\\ \mathbf{C}^{(k)}\mathbf{F}^{(k)}\mathbf{E}^{(k)^T} = \mathrm{SVD}\left(\mathbf{W}^{(k)}\right)\\ \mathbf{V}^{(k+1)} = \mathbf{C}^{(k)}\mathbf{E}^{(k)^T}.\end{cases} \quad (29)$$

5) Update $\mathcal{G}_1$: The $\mathcal{G}_1^{(k+1)}$ subproblem is deduced from (21) as

$$\mathcal{G}_1^{(k+1)} = \arg\min_{\mathcal{G}_1} \beta\Phi(\mathcal{G}_1) + \frac{\mu}{2}\left\|\nabla_1\mathcal{B}^{(k)} - \mathcal{G}_1 + \mathcal{H}_1^{(k)}\right\|_F^2. \quad (30)$$

According to the indicator function $\Phi(\cdot)$ defined in (6), then $\mathcal{G}_1^{(k+1)}$ is straightforwardly obtained as

$$\mathcal{G}_1^{(k+1)} = \mathcal{O}. \quad (31)$$

6) Update $\mathcal{G}_i$ (i = 2, 3): The $\mathcal{G}_i^{(k+1)}$ ($i = 2, 3$) subproblem is deduced from (21) as

$$\mathcal{G}_i^{(k+1)} = \arg\min_{\mathcal{G}_i} \beta\|\mathcal{G}_i\|_{\mathrm{TNN}} + \frac{\mu}{2}\left\|\nabla_i\mathcal{B}^{(k)} - \mathcal{G}_i + \mathcal{H}_i^{(k)}\right\|_F^2. \quad (32)$$

Through the tensor singular value thresholding scheme [33], $\mathcal{G}_i^{(k+1)}$ can be obtained as

$$\mathcal{G}_i^{(k+1)} = \mathrm{prox}_{\mathrm{TNN}}\left(\nabla_i\mathcal{B}^{(k)} + \mathcal{H}_i^{(k)}, \beta/\mu\right), \quad i = 2, 3. \quad (33)$$

7) Update $\mathcal{B}$: The $\mathcal{B}^{(k+1)}$ subproblem is deduced from (21) as

$$\mathcal{B}^{(k+1)} = \arg\min_{\mathcal{B}} \frac{1}{2}\left\|\mathcal{Y} - \mathcal{X}^{(k)} - \mathcal{B} - \mathcal{S}^{(k+1)}\right\|_F^2 + \frac{\mu}{2}\sum_{i=1}^{3}\left\|\nabla_i\mathcal{B} - \mathcal{G}_i^{(k+1)} + \mathcal{H}_i^{(k)}\right\|_F^2. \quad (34)$$

Evidently, it can be obtained by solving the following linear system:

$$\left(\mathcal{I} + \mu\sum_{i=1}^{3}\nabla_i^T\nabla_i\right)\mathcal{B} = \mathcal{Y} - \mathcal{X}^{(k)} - \mathcal{S}^{(k+1)} + \mu\sum_{i=1}^{3}\nabla_i^T\left(\mathcal{G}_i^{(k+1)} - \mathcal{H}_i^{(k)}\right) \quad (35)$$

where $\mathcal{I}$ is the identity operator.

Consequently, the fast Fourier transform is employed to efficiently solve (35), yielding $\mathcal{B}^{(k+1)}$ as

$$\mathcal{B}^{(k+1)} = \mathcal{F}^{-1}\left(\frac{\mathcal{F}(\mathcal{M} + \mu\mathcal{N})}{1 + \mu\sum_{i=1}^{3}|\mathcal{F}(\nabla_i)|^2}\right) \quad (36)$$

where $\mathcal{M} = \mathcal{Y} - \mathcal{X}^{(k)} - \mathcal{S}^{(k+1)}$, $\mathcal{N} = \sum_{i=1}^{3}\nabla_i^T(\mathcal{G}_i^{(k+1)} - \mathcal{H}_i^{(k)})$, $\mathcal{F}(\cdot)$ represents the 3-D fast Fourier transform, and $\mathcal{F}^{-1}(\cdot)$ corresponds to the inverse 3-D fast Fourier transform.

8) Update $\mathcal{X}$: The $\mathcal{X}^{(k+1)}$ subproblem is deduced from (21) as

$$\mathcal{X}^{(k+1)} = \arg\min_{\mathcal{X}} \frac{1}{2}\left\|\mathcal{Y} - \mathcal{X} - \mathcal{B}^{(k+1)} - \mathcal{S}^{(k+1)}\right\|_F^2 + \frac{\mu}{2}\left\|\nabla_3\mathcal{X} - \mathcal{U}^{(k+1)}\times_3\mathbf{V}^{(k+1)} + \mathcal{D}^{(k)}\right\|_F^2. \quad (37)$$

Likewise, by adopting the fast Fourier transform, we can straightforwardly solve $\mathcal{X}^{(k+1)}$ from (37) as

$$\mathcal{X}^{(k+1)} = \mathcal{F}^{-1}\left(\frac{\mathcal{F}(\mathcal{Q})}{1 + \mu|\mathcal{F}(\nabla_3)|^2}\right) \quad (38)$$

where $\mathcal{Q} = \mathcal{Y} - \mathcal{B}^{(k+1)} - \mathcal{S}^{(k+1)} + \mu\nabla_3^T(\mathcal{U}^{(k+1)}\times_3\mathbf{V}^{(k+1)} - \mathcal{D}^{(k)})$.

9) Update $\mathcal{D}$, $\mathcal{P}$, and $\mathcal{H}_i$: The Lagrange multipliers $\mathcal{D}^{(k+1)}$, $\mathcal{P}^{(k+1)}$, and $\mathcal{H}_i^{(k+1)}$ (i = 1, 2, 3) are updated by

$$\begin{cases}\mathcal{D}^{(k+1)} = \mathcal{D}^{(k)} + \nabla_3\mathcal{X}^{(k+1)} - \mathcal{U}^{(k+1)}\times_3\mathbf{V}^{(k+1)}\\ \mathcal{P}^{(k+1)} = \mathcal{P}^{(k)} + \mathcal{U}^{(k+1)} - \mathcal{A}^{(k+1)}\\ \mathcal{H}_i^{(k+1)} = \mathcal{H}_i^{(k)} + \nabla_i\mathcal{B}^{(k+1)} - \mathcal{G}_i^{(k+1)}.\end{cases} \quad (39)$$

To summarize, the complete optimization procedure for the proposed LLRSGSSTM model is described in Algorithm 1.

B. Theoretical Convergence Proof of LLRSGSSTM Algorithm

In addition, we give the theoretical proof of convergence of LLRSGSSTM algorithm in Theorem 1.

Theorem 1: If we denote the sequence produced by the LLRSGSSTM algorithm during the optimization process as $\{(\mathcal{X}^{(k)}, \mathcal{B}^{(k)}, \mathcal{S}^{(k)}, \mathcal{A}^{(k)}, \mathcal{G}_i^{(k)}, \mathcal{U}^{(k)}, \mathbf{V}^{(k)}, \mathcal{D}^{(k)}, \mathcal{P}^{(k)}, \mathcal{H}_i^{(k)})\}$, then it is bounded and its any accumulation point is a Karush–Kuhn–Tucker (KKT) point.

Proof: For the LLRSGSSTM algorithm, by leveraging the first-order optimality conditions of $\mathcal{X}^{(k)}$, $\mathcal{B}^{(k)}$, $\mathcal{S}^{(k)}$, $\mathcal{A}^{(k)}$, $\mathcal{G}_i^{(k)}$, $\mathcal{U}^{(k)}$, and $\mathbf{V}^{(k)}$ subproblems, we can get the relations

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Algorithm 1 LLRSGSSTM Algorithm

Input: The observed noisy HSI $\mathcal{Y} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, parameters λ, α, β, μ, the maximum number of iterations *MaxIter*, and the stopping criterion $\epsilon = 10^{-5}$.

Initialization: $k = 0$, $\mathcal{X}^{(0)} = \mathcal{Y}$, $\mathcal{S}^{(0)} = \mathcal{B}^{(0)} = \mathcal{G}_i^{(0)} = \mathcal{D}^{(0)} = \mathcal{H}_i^{(0)} = \mathcal{O} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, $i = 1, 2, 3$, $\mathcal{A}^{(0)} = \mathcal{U}^{(0)} = \mathcal{P}^{(0)} = \mathcal{O} \in \mathbb{R}^{n_1 \times n_2 \times r}$, $\mathbf{V}^{(0)} = \mathbf{O} \in \mathbb{R}^{n_3 \times r}$, where $\mathcal{O}$ represents the zero tensor, $\mathbf{O}$ represents the zero matrix.

Iterations:

For $k = 0$ **to** *MaxIter* $- 1$ **Do**
  Solve $\mathcal{S}^{(k+1)}$ via (23)
  Solve $\mathcal{A}^{(k+1)}$ via (25)
  Solve $\mathcal{U}^{(k+1)}$ via (27)
  Solve $\mathbf{V}^{(k+1)}$ via (29)
  Solve $\mathcal{G}_1^{(k+1)}$ via (31)
  Solve $\mathcal{G}_2^{(k+1)}$ and $\mathcal{G}_3^{(k+1)}$ via (33)
  Solve $\mathcal{B}^{(k+1)}$ via (36)
  Solve $\mathcal{X}^{(k+1)}$ via (38)
  Solve $\mathcal{D}^{(k+1)}$, $\mathcal{P}^{(k+1)}$ and $\mathcal{H}_i^{(k+1)}$ via (39)
  Checking the convergence rule $\|\mathcal{X}^{(k+1)} - \mathcal{X}^{(k)}\|_F / \|\mathcal{X}^{(k)}\|_F \le \epsilon$
End For

Output: Denoised HSI $\hat{\mathcal{X}}$.

between the subgradients $\partial(\cdot)$ and the Lagrange multipliers $(\mathcal{D}^{(k)}, \mathcal{P}^{(k)}, \mathcal{H}_i^{(k)})$ as

$$\begin{cases} \mu \nabla_3^T \mathcal{D}^{(k+1)} = \mathcal{Y} - \mathcal{X}^{(k+1)} - \mathcal{B}^{(k+1)} - \mathcal{S}^{(k+1)} \\ \mathcal{P}^{(k+1)} = \mathcal{D}^{(k+1)} \times_3 \mathbf{V}^{(k+1)^T} \\ \mu \mathcal{P}^{(k+1)} \in \partial\left(\alpha \mathrm{STV}_1\left(\mathcal{A}^{(k+1)}\right)\right) \\ \mu \mathcal{H}_1^{(k+1)} \in \partial\left(\beta \Phi\left(\mathcal{G}_1^{(k+1)}\right)\right) \\ \mu \mathcal{H}_i^{(k+1)} \in \partial\left(\beta \|\mathcal{G}_i^{(k+1)}\|_{\mathrm{TNN}}\right), \quad i = 2, 3 \\ \mu \sum_{i=1}^{3} \nabla_i^T \mathcal{H}_i^{(k+1)} = \mathcal{Y} - \mathcal{X}^{(k+1)} - \mathcal{B}^{(k+1)} - \mathcal{S}^{(k+1)}. \end{cases} \tag{40}$$

Relied on [59, Th. 4 and Lemma 1], we get that $(\mathcal{D}^{(k)}, \mathcal{P}^{(k)}, \mathcal{H}_i^{(k)})$ is bounded, $\|\mathcal{D}^{(k)} - \mathcal{D}^{(k-1)}\|_F^2 \to 0$, $\|\mathcal{P}^{(k)} - \mathcal{P}^{(k-1)}\|_F^2 \to 0$, and $\|\mathcal{H}_i^{(k)} - \mathcal{H}_i^{(k-1)}\|_F^2 \to 0$ when $k \to \infty$. Furthermore, relied on the bounded property of $(\mathcal{D}^{(k)}, \mathcal{P}^{(k)}, \mathcal{H}_i^{(k)})$, and the coercive property of $\|\cdot\|_F^2$, $\|\cdot\|_1$, $\mathrm{STV}_1(\cdot)$, $\Phi(\cdot)$, and $\|\cdot\|_{\mathrm{TNN}}$ in the objective function of LLRSGSSTM model (20), we get that $\{(\mathcal{X}^{(k)}, \mathcal{B}^{(k)}, \mathcal{S}^{(k)}, \mathcal{A}^{(k)}, \mathcal{G}_i^{(k)}, \mathcal{U}^{(k)}, \mathbf{V}^{(k)})\}$ is bounded. Therefore, the whole sequence $\{(\mathcal{X}^{(k)}, \mathcal{B}^{(k)}, \mathcal{S}^{(k)}, \mathcal{A}^{(k)}, \mathcal{G}_i^{(k)}, \mathcal{U}^{(k)}, \mathbf{V}^{(k)}, \mathcal{D}^{(k)}, \mathcal{P}^{(k)}, \mathcal{H}_i^{(k)})\}$ is bounded.

Meanwhile, relied on the Bolzano–Weierstrass theorem, $\{(\mathcal{X}^{(k)}, \mathcal{B}^{(k)}, \mathcal{S}^{(k)}, \mathcal{A}^{(k)}, \mathcal{G}_i^{(k)}, \mathcal{U}^{(k)}, \mathbf{V}^{(k)}, \mathcal{D}^{(k)}, \mathcal{P}^{(k)}, \mathcal{H}_i^{(k)})\}$ has at least one accumulation point $(\hat{\mathcal{X}}, \hat{\mathcal{B}}, \hat{\mathcal{S}}, \hat{\mathcal{A}}, \hat{\mathcal{G}}_i, \hat{\mathcal{U}}, \hat{\mathbf{V}}, \hat{\mathcal{D}}, \hat{\mathcal{P}}, \hat{\mathcal{H}}_i)$. Without loss of generality, we denote $\{(\mathcal{X}^{(k)}, \mathcal{B}^{(k)}, \mathcal{S}^{(k)}, \mathcal{A}^{(k)}, \mathcal{G}_i^{(k)}, \mathcal{U}^{(k)}, \mathbf{V}^{(k)}, \mathcal{D}^{(k)}, \mathcal{P}^{(k)}, \mathcal{H}_i^{(k)})\}$ as the subsequence converging to the accumulation point $(\hat{\mathcal{X}}, \hat{\mathcal{B}}, \hat{\mathcal{S}}, \hat{\mathcal{A}}, \hat{\mathcal{G}}_i, \hat{\mathcal{U}}, \hat{\mathbf{V}}, \hat{\mathcal{D}}, \hat{\mathcal{P}}, \hat{\mathcal{H}}_i)$ when $k \to \infty$.

Specifically, relied on the closed property of subgradients $\partial(\cdot)$, the accumulation point $(\hat{\mathcal{X}}, \hat{\mathcal{B}}, \hat{\mathcal{S}}, \hat{\mathcal{A}}, \hat{\mathcal{G}}_i, \hat{\mathcal{U}}, \hat{\mathbf{V}}, \hat{\mathcal{D}}, \hat{\mathcal{P}}, \hat{\mathcal{H}}_i)$ clearly satisfies

$$\begin{cases} 0 \in \partial\left(\lambda \|\hat{\mathcal{S}}\|_1\right) + \partial\left(\alpha \, \mathrm{STV}_1\left(\hat{\mathcal{A}}\right)\right) \\ \qquad + \partial\left(\beta \Phi\left(\hat{\mathcal{G}}_1\right)\right) + \partial\left(\beta \sum_{i=2}^{3} \|\hat{\mathcal{G}}_i\|_{\mathrm{TNN}}\right) \\ \qquad + 3\left(\hat{\mathcal{X}} + \hat{\mathcal{B}} + \hat{\mathcal{S}} - \mathcal{Y}\right) \\ 0 = \nabla_3 \hat{\mathcal{X}} - \hat{\mathcal{U}} \times_3 \hat{\mathbf{V}} \\ 0 = \hat{\mathbf{V}}^T \hat{\mathbf{V}} - \mathbf{I} \\ 0 = \hat{\mathcal{U}} - \hat{\mathcal{A}} \\ 0 = \nabla_i \hat{\mathcal{B}} - \hat{\mathcal{G}}_i, \quad i = 1, 2, 3. \end{cases} \tag{41}$$

Thus, we get that $(\hat{\mathcal{X}}, \hat{\mathcal{B}}, \hat{\mathcal{S}}, \hat{\mathcal{A}}, \hat{\mathcal{G}}_i, \hat{\mathcal{U}}, \hat{\mathbf{V}}, \hat{\mathcal{D}}, \hat{\mathcal{P}}, \hat{\mathcal{H}}_i)$ is a KKT point of the LLRSGSSTM model, which completes the convergence proof of LLRSGSSTM algorithm.

IV. Experiments

A. Experimental Settings

To comprehensively assess the denoising performance of LLRSGSSTM, a series of experiments is conducted on simulated and real hyperspectral datasets. Specifically, simulated denoising experiments are carried out on the Houston and Pavia City Center (PaviaC) datasets, while real-data experiments are performed on the real GaoFen-5 (GF-5) Baoqing and Earth Observing-1 (EO-1) Hyperion [60] datasets.

For the performance comparison, the proposed LLRSGSSTM is compared against some state-of-the-art HSI denoising methods, i.e., LRTV [39], 3DTNN [33], E-3DTV [52], RCTV [49], 3DCTV [46], t-CTV [47], fast graph Laplacian regularizer (FGLR) [31], HLTVSG [53], and fast HSI mixed noise removal (FastHyMix) [10].

Prior to denoising, all hyperspectral datasets are normalized band-wise to the range of [0, 1] to ensure a fair comparison. For the simulated experiments, the quantitative performance is assessed by using three common metrics, i.e., spectral angle mapper (SAM), mean peak signal-to-noise ratio (MPSNR), and mean structural similarity (MSSIM).

Moreover, all denoising algorithms are implemented in the MATLAB R2022a environment on a workstation with the Windows 11 operating system, the Intel Core i7-10875H CPU (2.30 GHz), and 16 GB of RAM.

B. Simulated Denoising Experiments

1) Simulation Setting: The simulated denoising experiments are performed using the Houston ($200 \times 200 \times 144$) and PaviaC ($200 \times 200 \times 80$) datasets. To investigate the denoising behavior of different methods under mixed noise scenarios, four simulation cases are designed by contaminating the clean HSI with various types of noise.

Case 1 (Gaussian Noise + Stripe Noise): First, Gaussian noise is added to each spectral band, with the noise standard deviation randomly selected from the range of [0.1, 0.15]. Subsequently, stripe noise is randomly introduced into 60% of the spectral bands, and within each selected band, 30% of the columns are corrupted by stripe noise.

Case 2 (Gaussian Noise + Stripe Noise + Impulse Noise): The Gaussian and stripe noise settings remain the same

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TABLE I
QUANTITATIVE COMPARISONS OF MPSNR, MSSIM, AND SAM ON THE SIMULATED HOUSTON AND PAVIAC DATASETS

Datasets	Cases	Indexes	Noisy	LRTV	3DTNN	E-3DTV	RCTV	3DCTV	t-CTV	FGLR	HLTVSG	FastHyMix	LLRSGSSTM
Houston	Case 1	MPSNR	18.39	25.82	27.42	27.96	27.46	27.50	25.69	29.01	28.57	31.20	**31.28**
		MSSIM	0.2236	0.6366	0.7044	0.8000	0.7544	0.7476	0.5806	0.7978	0.8011	0.8761	**0.8846**
		SAM	29.7751	12.2669	10.7642	8.4027	11.1466	10.3912	13.3501	8.5999	9.5087	7.0204	**6.5161**
	Case 2	MPSNR	18.26	25.76	27.31	27.95	27.25	27.32	25.49	28.94	28.62	30.57	**30.71**
		MSSIM	0.2193	0.6346	0.6960	0.7971	0.7477	0.7364	0.5695	0.7947	0.7979	0.8634	**0.8747**
		SAM	30.0750	12.9191	11.4441	8.7290	12.0026	11.1215	14.0429	9.0193	9.5523	8.0947	**7.2879**
	Case 3	MPSNR	18.28	25.82	27.27	27.95	27.44	27.44	25.50	29.06	28.61	31.16	**31.17**
		MSSIM	0.2196	0.6290	0.6956	0.7958	0.7489	0.7377	0.5716	0.7976	0.7989	0.8673	**0.8812**
		SAM	30.2982	12.7702	11.4778	8.3863	11.3957	10.6142	14.0196	8.6056	9.3620	7.3463	**6.6705**
	Case 4	MPSNR	18.20	25.70	27.14	27.97	27.30	27.30	25.40	28.97	28.56	30.62	**30.67**
		MSSIM	0.2175	0.6308	0.6877	0.8027	0.7475	0.7325	0.5676	0.7947	0.7964	0.8543	**0.8759**
		SAM	30.3133	13.0232	11.8466	9.0871	11.9137	11.3283	14.3755	9.0370	9.8540	8.0123	**7.4417**
PaviaC	Case 1	MPSNR	18.24	24.98	26.80	27.05	26.64	27.03	25.07	28.38	27.99	30.77	**30.86**
		MSSIM	0.3093	0.6306	0.7711	0.7997	0.7695	0.7843	0.6586	0.8452	0.8301	0.9031	**0.9067**
		SAM	28.6551	11.2238	8.1241	6.7215	8.8785	8.7235	12.1074	6.6004	7.0286	6.2537	**5.0384**
	Case 2	MPSNR	18.23	24.69	26.85	26.71	26.64	26.97	25.05	28.41	28.16	30.59	**30.63**
		MSSIM	0.3084	0.6345	0.7698	0.7933	0.7742	0.7860	0.6583	0.8488	0.8361	0.9020	**0.9046**
		SAM	28.4475	11.6273	8.2657	6.8862	9.1476	8.9460	12.1841	6.6288	6.7575	6.2666	**5.0676**
	Case 3	MPSNR	18.05	25.07	26.43	27.19	26.57	26.91	24.66	28.37	28.08	30.44	**30.68**
		MSSIM	0.3038	0.6534	0.7370	0.8185	0.7768	0.7852	0.6432	0.8472	0.8337	0.8919	**0.9038**
		SAM	29.0690	11.1876	9.7813	6.9977	9.5687	9.2090	13.2464	6.7835	6.9544	6.9615	**5.2620**
	Case 4	MPSNR	18.13	24.97	26.60	27.13	26.66	26.96	24.78	28.44	28.00	30.34	**30.67**
		MSSIM	0.3050	0.6350	0.7553	0.8122	0.7737	0.7847	0.6492	0.8481	0.8334	0.8926	**0.9046**
		SAM	28.6264	11.2666	9.1337	6.5251	9.1798	8.9255	12.9007	6.6023	6.8128	6.6908	**4.8407**



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k) (l)

Fig. 3. Comparison of restored HSIs generated by different methods on band 93 of the simulated Houston dataset in Case 4. (a) Original HSI, (b) noisy HSI, (c) LRTV, (d) 3DTNN, (e) E-3DTV, (f) RCTV, (g) 3DCTV, (h) t-CTV, (i) FGLR, (j) HLTVSG, (k) FastHyMix, and (l) LLRSGSSTM.

as those in Case 1. Besides, impulse noise with a density randomly selected from the range of [0, 0.01] is injected into 20% of the spectral bands.

Case 3 (Gaussian Noise + Stripe Noise + Deadlines): In Case 3, the same Gaussian and stripe noise as in Case 1 are added. In addition, deadlines are imposed on 11–20 spectral bands that contain stripe noise. For every selected band, the number of deadlines is randomly chosen from [2, 5], and the width of each deadline is randomly selected from [1, 3].

Case 4 (Gaussian Noise + Stripe Noise + Impulse Noise + Deadlines): In this case, the noisy HSI is generated by further adding impulse noise with the same setting as in Case 2 to the degraded data in Case 3.

2) Quantitative and Visual Comparisons: Table I describes the quantitative denoising results of MPSNR, MSSIM, and SAM, where the best and second-best performance are bolded and underlined. Specifically, LLRSGSSTM consistently achieves the highest MPSNR and MSSIM values, as well as the lowest SAM values, across all simulation cases on both HSI datasets, which indicates that LLRSGSSTM is more effective in suppressing mixed noise, particularly stripe-related degradations.

For qualitative visual evaluation, Case 4 is selected as a representative scenario, as it involves the most severe and complex mixed noise conditions. Figs. 3–6 present the visual denoising results and the corresponding spectral curves of the simulated Houston and PaviaC HSIs in Case 4, including the noisy observations and the restored results.

From the visual comparisons in Figs. 3 and 5, LRTV, E-3DTV, and RCTV tend to produce noticeable oversmoothing effects, leading to the loss of fine spatial structures and image details. In contrast, although 3DTNN, 3DCTV, t-CTV, FGLR,

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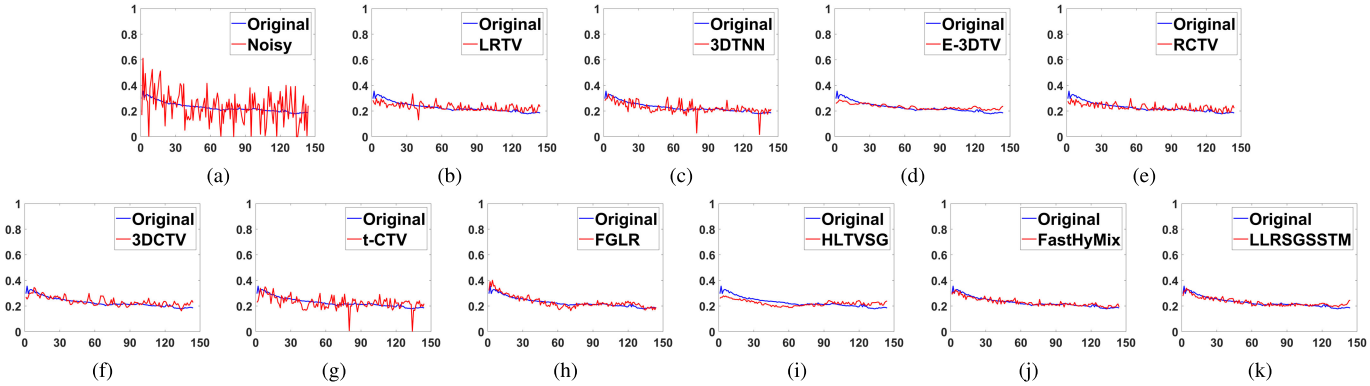


Fig. 4. Comparison of spectral curves generated by different methods at spatial pixels (65, 172) on the simulated Houston dataset in Case 4. (a) Noisy HSI, (b) LRTV, (c) 3DTNN, (d) E-3DTV, (e) RCTV, (f) 3DCTV, (g) t-CTV, (h) FGLR, (i) HLTVSG, (j) FastHyMix, and (k) LLRSGSSTM.

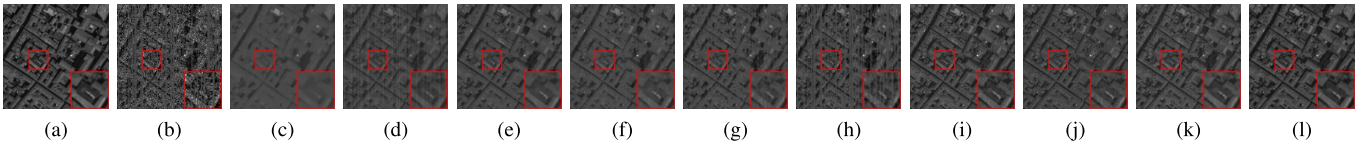


Fig. 5. Comparison of restored HSIs generated by different methods on band 45 of the simulated PaviaC dataset in Case 4. (a) Original HSI, (b) noisy HSI, (c) LRTV, (d) 3DTNN, (e) E-3DTV, (f) RCTV, (g) 3DCTV, (h) t-CTV, (i) FGLR, (j) HLTVSG, (k) FastHyMix, and (l) LLRSGSSTM.

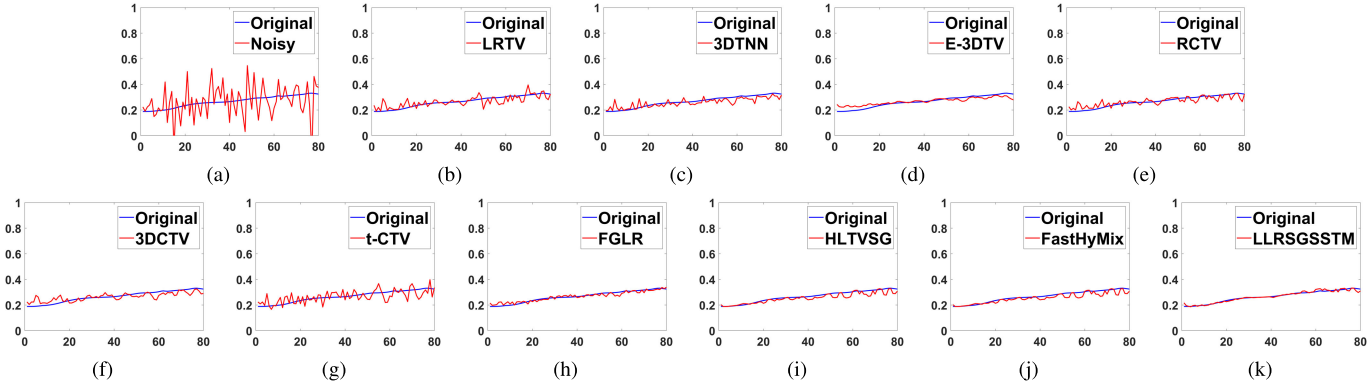


Fig. 6. Comparison of spectral curves generated by different methods at spatial pixels (65, 35) on the simulated PaviaC dataset in Case 4. (a) Noisy HSI, (b) LRTV, (c) 3DTNN, (d) E-3DTV, (e) RCTV, (f) 3DCTV, (g) t-CTV, (h) FGLR, (i) HLTVSG, (j) FastHyMix, and (k) LLRSGSSTM.

HLTVSG, and FastHyMix preserve more image details, they suffer from evident residual noise, especially stripe noise, which remains visible in the restored images. For comparison, LLRSGSSTM demonstrates superior visual denoising and destriping performance, where most mixed noise is more effectively removed, while important structural information of the HSIs is better preserved.

As shown in Figs. 4 and 6, the spectral curves provide an intuitive comparison of how different methods affect spectral consistency after denoising. It can be observed that several competing methods alter the spectral trends in local bands, leading to discontinuities or inconsistent responses along the spectral dimension. In contrast, the spectral curves restored by LLRSGSSTM follow the reference curves with stable transitions between adjacent bands, indicating that the spectral correlation is better preserved during noise removal. Specifically, LLRSGSSTM effectively suppresses mixed noise without disrupting the inherent spectral information, thereby preserving faithful spectral information in the restored HSIs.

Furthermore, Fig. 7 shows the band-wise PSNR and SSIM curves of different denoising approaches on Houston and PaviaC datasets in Case 4. Most existing approaches exhibit noticeable fluctuations across spectral bands, where the performance drops in those bands heavily affected by mixed and stripe noise. While LRTV, 3DTNN, E-3DTV, RCTV, 3DCTV, t-CTV, FGLR, HLTVSG, and FastHyMix achieve acceptable results in cleaner bands, their curves remain uneven, reflecting limited robustness to band-dependent noise variations. In contrast, LLRSGSSTM maintains consistently high PSNR and SSIM values across nearly all bands, with much smoother behavior along the spectral dimension. Specifically, this cross-band stability is particularly evident in the PaviaC dataset, highlighting the ability of LLRSGSSTM to provide reliable denoising performance even when noise characteristics vary significantly between bands.

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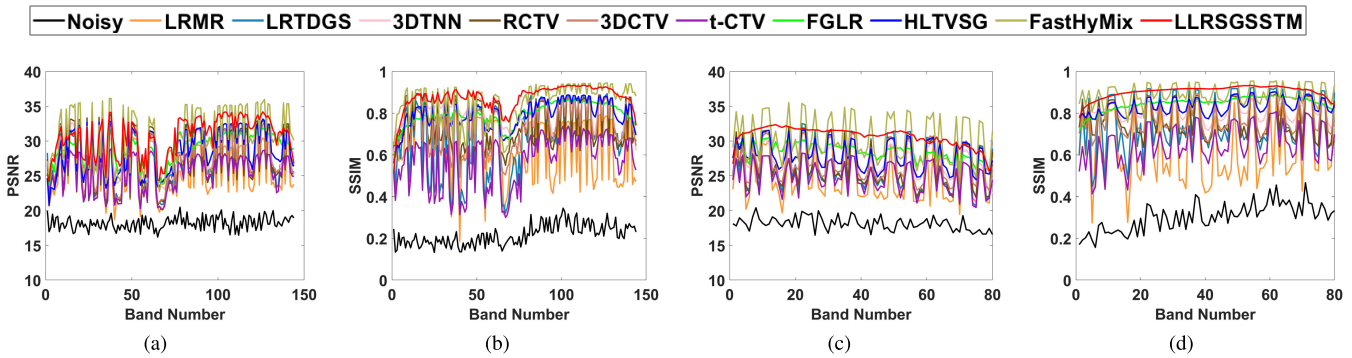


Fig. 7. Comparison of band-wise PSNR and SSIM values achieved by different methods on simulated Houston and PaviaC datasets in Case 4. (a) PSNR of each spectral band on Houston, (b) SSIM of each spectral band on Houston, (c) PSNR of each spectral band on PaviaC, and (d) SSIM of each spectral band on PaviaC.

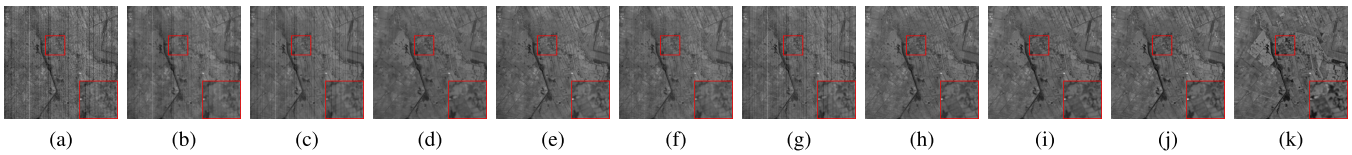


Fig. 8. Comparison of restored HSIs generated by different methods on band 154 of the real GF-5 Baoqing dataset. (a) Real noisy HSI, (b) LRTV, (c) 3DTNN, (d) E-3DTV, (e) RCTV, (f) 3DCTV, (g) t-CTV, (h) FGLR, (i) HLTVSG, (j) FastHyMix, and (k) LLRSGSSTM.

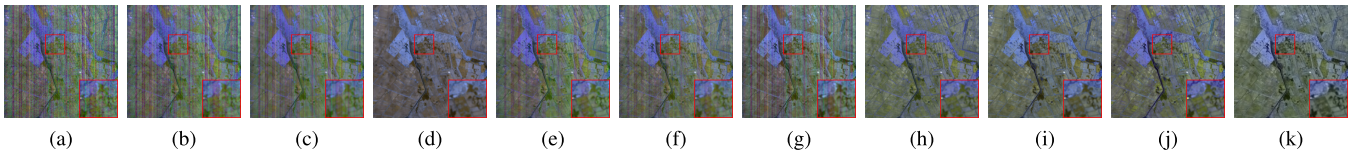


Fig. 9. Comparison of pseudocolor restored HSIs generated by different methods on bands (154, 155, 21) of the real GF-5 Baoqing dataset. (a) Real noisy HSI, (b) LRTV, (c) 3DTNN, (d) E-3DTV, (e) RCTV, (f) 3DCTV, (g) t-CTV, (h) FGLR, (i) HLTVSG, (j) FastHyMix, and (k) LLRSGSSTM.

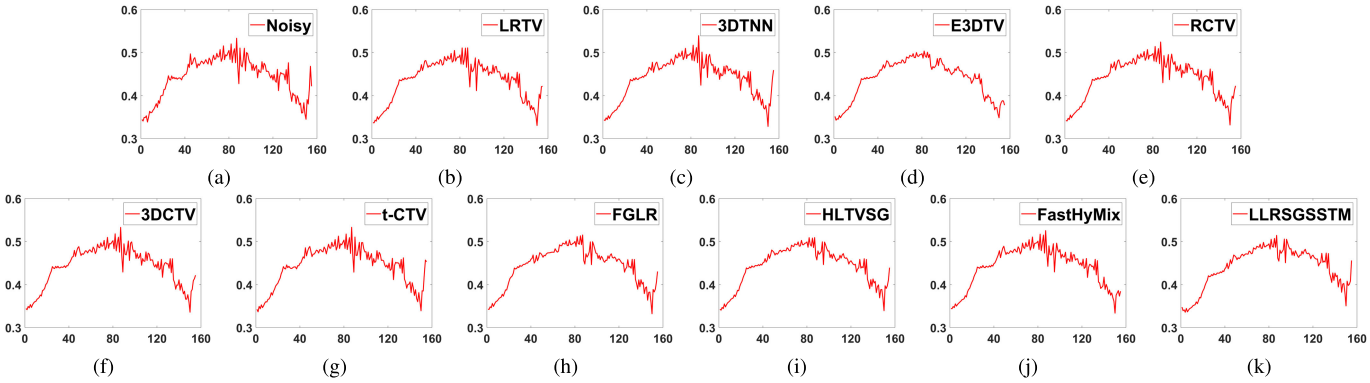


Fig. 10. Comparison of spectral curves generated by different methods at spatial pixels (105, 163) on the real GF-5 Baoqing dataset. (a) Real noisy HSI, (b) LRTV, (c) 3DTNN, (d) E-3DTV, (e) RCTV, (f) 3DCTV, (g) t-CTV, (h) FGLR, (i) HLTVSG, (j) FastHyMix, and (k) LLRSGSSTM.

C. Real Denoising Experiments

For the real-data experiments, in Figs. 8–15, we assess the performance of different methods on the GF-5 Baoqing dataset ($300 \times 300 \times 155$) and the EO-1 Hyperion dataset ($200 \times 200 \times 166$), where multiple spectral bands are seriously affected by mixed noise and stripe noise, making these datasets more challenging for evaluating noise removal effectiveness.

The visual results on the GF-5 Baoqing and EO-1 Hyperion datasets, as shown in Figs. 8 and 12, reveal differences in how existing methods handle real noise. LRTV, 3DTNN, E-3DTV, RCTV, 3DCTV, t-CTV, FGLR, HLTVSG, and FastHyMix all demonstrate varying degrees of residual stripe patterns or incomplete removal of complex noise, particularly in regions heavily affected by mixed noise. Some methods leave subtle stripes in uniform areas, while others blur fine textures, indicating challenges in simultaneously preserving spatial details and suppressing noise. In contrast, LLRSGSSTM produces cleaner images with consistent spatial structures, effectively reducing both stripe artifacts and mixed noise while maintaining the integrity of edges and textures throughout the scene.

Figs. 9 and 13 further display the pseudocolor restoration results to assess the spectral–spatial recovery performance.

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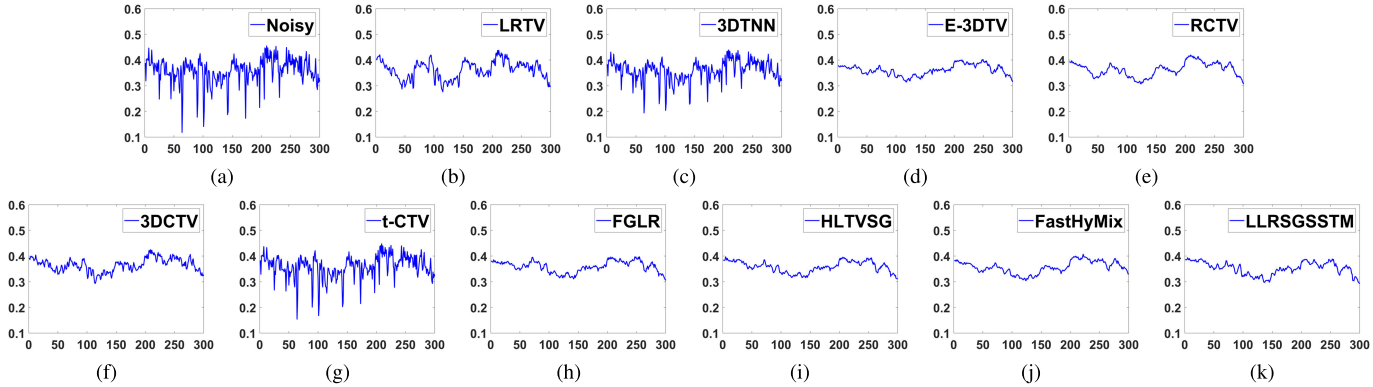


Fig. 11. Column mean curves obtained by various denoising methods on band 152 of the real GF-5 Baoqing dataset. (a) Real noisy HSI, (b) LRTV, (c) 3DTNN, (d) E-3DTV, (e) RCTV, (f) 3DCTV, (g) t-CTV, (h) FGLR, (i) HLTVSG, (j) FastHyMix, and (k) LLRSGSSTM.

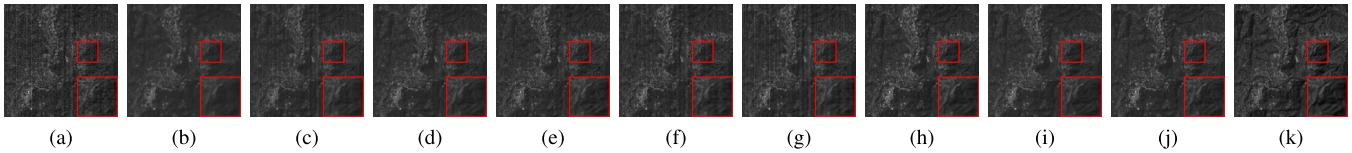


Fig. 12. Comparison of restored HSIs generated by different methods on band 164 of the real EO-1 Hyperion dataset. (a) Real noisy HSI, (b) LRTV, (c) 3DTNN, (d) E-3DTV, (e) RCTV, (f) 3DCTV, (g) t-CTV, (h) FGLR, (i) HLTVSG, (j) FastHyMix, and (k) LLRSGSSTM.

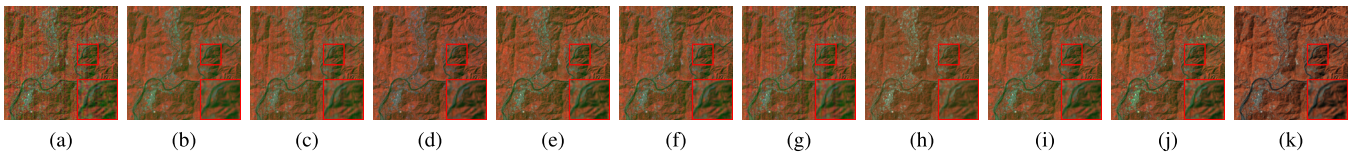


Fig. 13. Comparison of pseudocolor restored HSIs generated by different methods on bands (33, 166, and 150) of the real EO-1 Hyperion dataset. (a) Real noisy HSI, (b) LRTV, (c) 3DTNN, (d) E-3DTV, (e) RCTV, (f) 3DCTV, (g) t-CTV, (h) FGLR, (i) HLTVSG, (j) FastHyMix, and (k) LLRSGSSTM.

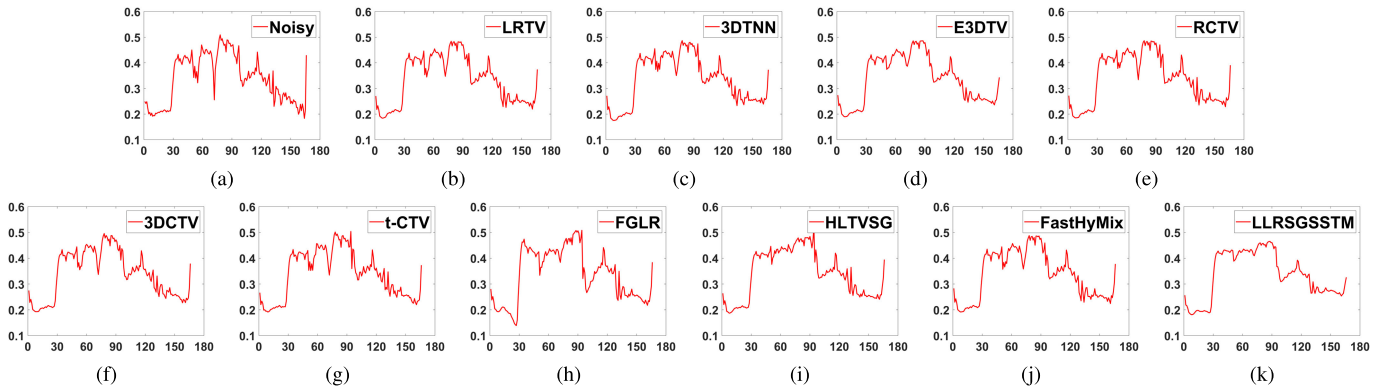


Fig. 14. Comparison of spectral curves generated by different methods at spatial pixels (182, 45) on the real EO-1 Hyperion dataset. (a) Real noisy HSI, (b) LRTV, (c) 3DTNN, (d) E-3DTV, (e) RCTV, (f) 3DCTV, (g) t-CTV, (h) FGLR, (i) HLTVSG, (j) FastHyMix, and (k) LLRSGSSTM.

As observed from Figs. 9 and 13, LRTV, 3DTNN, E-3DTV, RCTV, 3DCTV, t-CTV, FGLR, HLTVSG, and FastHyMix often suffer from obvious residual noise or color distortion. On the contrary, LLRSGSSTM can more effectively eliminate the serious mixed noise while keeping the natural colors, which demonstrates the strong spatial–spectral recovery ability of LLRSGSSTM.

Meanwhile, to assess the spectral fidelity, Figs. 10 and 14 display the spectral curves of selected spatial pixels. Due to the adverse influence of residual noise or oversmoothing, the compared denoising approaches often produce curves with apparent oscillations. Specifically, LLRSGSSTM generates highly smooth spectral profiles across all the spectral bands.

Finally, the column mean curves of several heavily contaminated bands are depicted in Figs. 11 and 15 to examine the effectiveness of noise suppression on real data. From these curves, it is evident that residual fluctuations remain after denoising for many approaches, with irregular variations indicating that stripe noise and other structured disturbances are not adequately removed. In contrast, LLRSGSSTM yields

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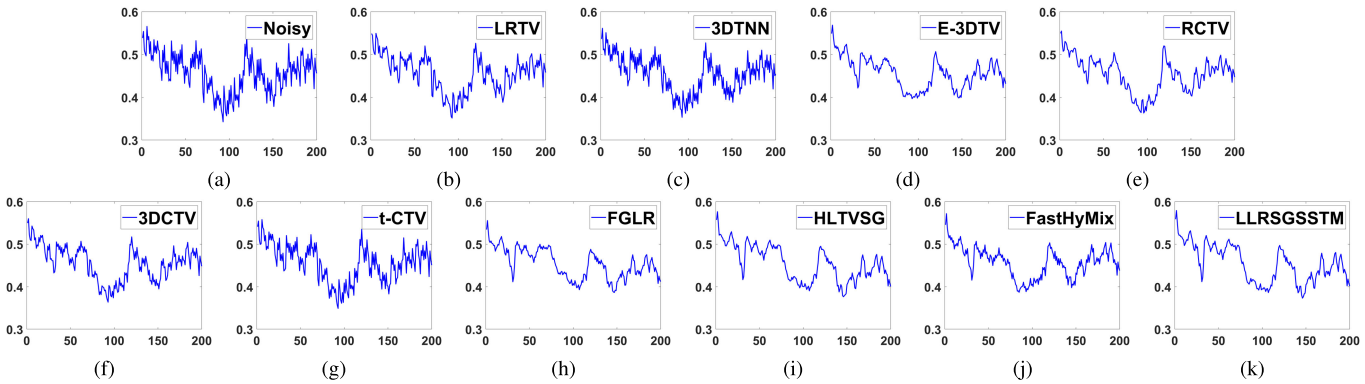


Fig. 15. Column mean curves obtained by various denoising methods on band 53 of the real EO-1 Hyperion dataset. (a) Real noisy HSI, (b) LRTV, (c) 3DTNN, (d) E-3DTV, (e) RCTV, (f) 3DCTV, (g) t-CTV, (h) FGLR, (i) HLTVSG, (j) FastHyMix, and (k) LLRSGSSTM.

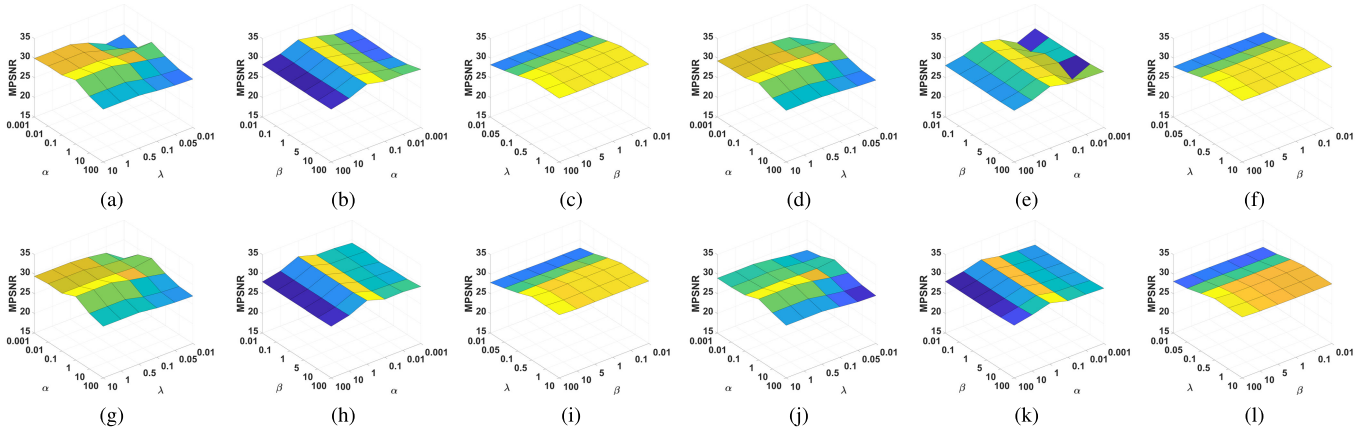


Fig. 16. Sensitivity of regularization parameters λ, α, and β for LLRSGSSTM on the simulated Houston dataset in Cases 1–4. (a)–(c) MPSNR versus (λ,α), (α,β), and (β,λ) in Case 1. (d)–(f) MPSNR versus (λ,α), (α,β), and (β,λ) in Case 2. (g)–(i) MPSNR versus (λ,α), (α,β), and (β,λ) in Case 3. (j)–(l) MPSNR versus (λ,α), (α,β), and (β,λ) in Case 4.

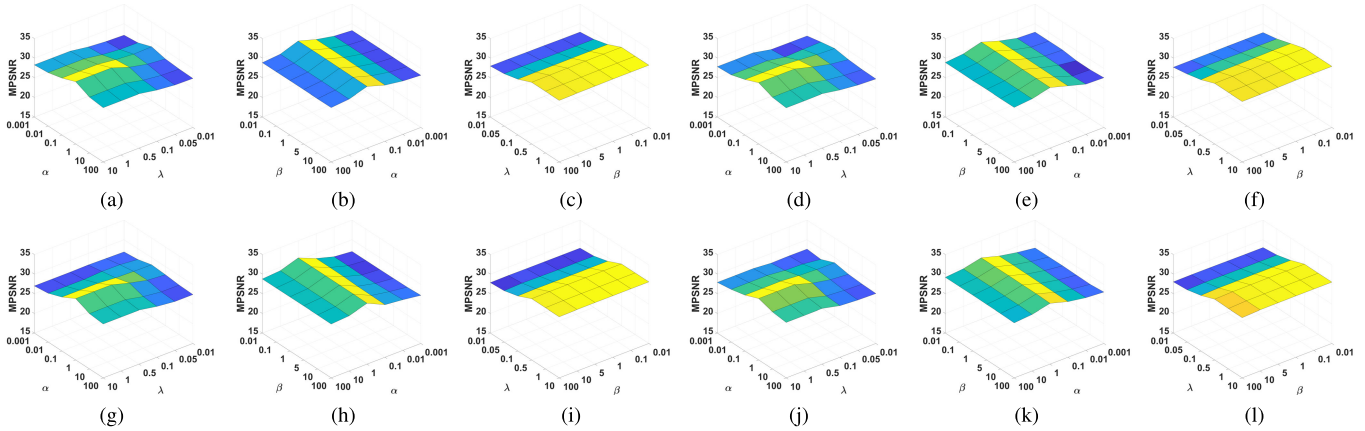


Fig. 17. Sensitivity of regularization parameters λ, α, and β for LLRSGSSTM on the simulated PaviaC dataset in Cases 1–4. (a)–(c) MPSNR versus (λ,α), (α,β), and (β,λ) in Case 1. (d)–(f) MPSNR versus (λ,α), (α,β), and (β,λ) in Case 2. (g)–(i) MPSNR versus (λ,α), (α,β), and (β,λ) in Case 3. (j)–(l) MPSNR versus (λ,α), (α,β), and (β,λ) in Case 4.

column mean curves that are markedly smoother and more continuous, exhibiting reduced oscillations and a clearer global trend. Such behavior implies that LLRSGSSTM is better able to recover the latent image characteristics from severely degraded observations, leading to more stable denoising results on real HSIs.

D. Parameter Analysis

1) Sensitivity of Regularization Parameters λ, α, and β: For LLRSGSSTM, regularization parameters λ, α, and β play essential roles in determining the overall effectiveness of denoising and destriping. To evaluate their influence, Figs. 16 and 17 present the MPSNR results on the simulated Houston

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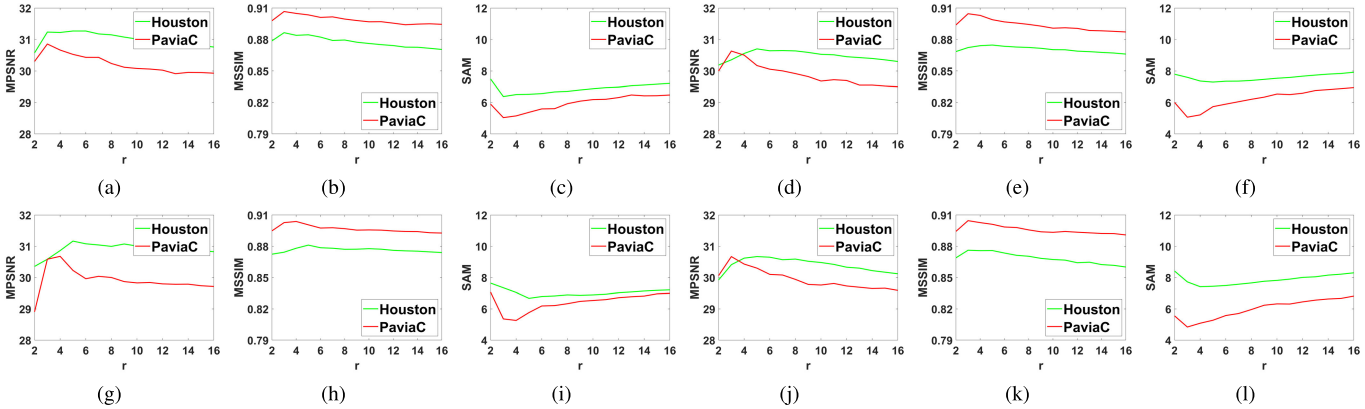


Fig. 18. Parameter sensitivity of the SGS dimension r for LLRSGSSTM on the simulated Houston and PaviaC datasets in Cases 1–4. (a)–(c) MPSNR, MSSIM, and SAM versus r in Case 1. (d)–(f) MPSNR, MSSIM, and SAM versus r in Case 2. (g)–(i) MPSNR, MSSIM, and SAM versus r in Case 3. (j)–(l) MPSNR, MSSIM, and SAM versus r in Case 4.

and PaviaC datasets in Cases 1–4 under various parameter settings. Evidently, different settings of λ, α, and β lead to noticeable differences in the quality of restored images. Through comprehensive empirical testing, the optimal parameter ranges across these cases are found to be $\lambda \in [0.1, 0.5]$, $\alpha \in [1, 10]$, and $\beta \in [10, 100]$ for the Houston dataset. Meanwhile, the ranges of $\lambda \in [0.1, 0.5]$, $\alpha \in [0.5, 1.5]$, and $\beta \in [0.1, 5]$ achieve a favorable balance for the PaviaC dataset.

2) Sensitivity of SGS Dimension r: To further investigate the influence of the SGS dimension r in LLRSGSSTM, we evaluate its effect on the MPSNR, MSSIM, and SAM metrics by using the simulated Houston and PaviaC datasets in Cases 1–4. As depicted in Fig. 18, the selection of r has a noticeable impact on the overall restoration performance. Based on our empirical analysis, setting $r = 5$ for the Houston dataset and $r = 3$ for the PaviaC dataset yields optimal denoising and destriping results.

E. Model Ablation Analysis

For the unified multimode gradient tensor prior term of stripe noise $\mathcal{B}$ in (9) and the SGSRC-based STV prior of HSI $\mathcal{X}$ in (18), we aim at studying how these two prior terms affect the overall denoising performance of the proposed LLRSGSSTM model (19). Thus, the following two submodels are analyzed.

1) *Submodel 1*: LLRSGSSTM without the SGSRC-based STV prior of HSI $\mathcal{X}$ in (18), i.e., LLRSGSSTM with $\alpha = 0$.
2) *Submodel 2*: LLRSGSSTM without the unified multimode gradient tensor prior term of stripe noise $\mathcal{B}$ in (9), i.e., LLRSGSSTM with $\beta = 0$.

Moreover, Table II describes the quantitative SAM, MPSNR, and MSSIM results, while Fig. 19 provides the corresponding visually restored HSIs of LLRSGSSTM and its aforementioned two submodels (i.e., LLRSGSSTM with $\alpha = 0$ and LLRSGSSTM with $\beta = 0$) on the simulated PaviaC dataset in Case 4.

Apparently, in Table II and Fig. 19, the whole LLRSGSSTM model shows the best SAM, MPSNR, and MSSIM coupled with the most excellent visual restoration quality for HSI destriping and denoising, where either LLRSGSSTM with $\alpha = 0$ or LLRSGSSTM with $\beta = 0$ shows much less SAM, MPSNR, and MSSIM results and also shows more visual artifacts. Thus, we obtain that either the unified multimode gradient tensor prior term of stripe noise $\mathcal{B}$ in (9) or the SGSRC-based STV prior of HSI $\mathcal{X}$ in (18) plays a complementary yet essential role in the LLRSGSSTM model (19).

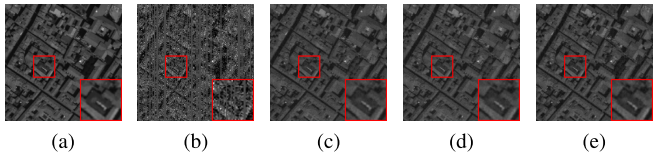


Fig. 19. Comparison of restored HSIs generated by different submodels on band 45 of the simulated PaviaC dataset in Case 4. (a) Original HSI, (b) noisy HSI, (c) submodel 1, (d) submodel 2, and (e) LLRSGSSTM.

TABLE II
MODEL ABLATION STUDY OF LLRSGSSTM ON THE SIMULATED PAVIAC DATASET IN CASE 4

	MPSNR	MSSIM	SAM
Sub-model 1: LLRSGSSTM with $\alpha = 0$	24.98	0.8449	10.9990
Sub-model 2: LLRSGSSTM with $\beta = 0$	28.35	0.8681	6.9918
LLRSGSSTM	**30.67**	**0.9046**	**4.8407**

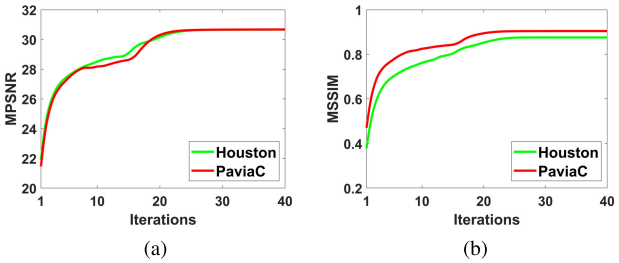


Fig. 20. Analysis of convergence curves for LLRSGSSTM on simulated Houston and PaviaC datasets in Case 4. (a) MPSNR versus iterations and (b) MSSIM versus iterations.

F. Convergence Analysis

To validate the optimization stability of the proposed LLRSGSSTM algorithm summarized in Algorithm 1, we thoroughly

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TABLE III
COMPARISON OF RUNNING TIMES (IN SECONDS) OF DIFFERENT DENOISING APPROACHES ON THE SIMULATED AND REAL DATASETS

Datasets	Size	LRTV	3DTNN	E-3DTV	RCTV	3DCTV	t-CTV	FGLR	HLTVSG	FastHyMix	LLRSGSSTM
Houston	$200 \times 200 \times 144$	78.6	104.3	22.2	5.4	36.9	242.5	2.9	36.6	0.2	120.8
PaviaC	$200 \times 200 \times 80$	38.6	50.5	12.5	3.5	17.1	139.1	1.6	25.6	0.2	90.2
GF-5 Baoqing	$300 \times 300 \times 155$	184.2	132.8	53.6	13.4	93.1	629.7	4.0	60.2	0.6	338.9
EO-1 Hyperion	$200 \times 200 \times 166$	71.7	77.6	26.6	5.9	45.1	271.3	2.3	38.3	0.3	109.1

investigate its convergence behavior. Fig. 20 plots the progressive evolution of MPSNR and MSSIM metrics versus the iteration number on simulated Houston and PaviaC datasets in Case 4. Visually, the MSPNR and MSSIM results increase rapidly at earlier iterations and progressively converge to a steady state. Such a progressive optimization trajectory clearly demonstrates that LLRSGSSTM can effectively and continuously refine the restored HSI, guaranteeing a reliable convergence to a stable high-quality solution.

In light of these observations, the maximum number of iterations is empirically set as $MaxIter = 40$ for all experimental setups. The plotted convergence trajectories robustly verify that this predefined termination condition is fully sufficient to ensure the complete convergence of the proposed LLRSGSSTM algorithm.

G. Running Time Analysis

Table III further describes the running time of the evaluated methods on the simulated Houston and PaviaC datasets, and real GF-5 Baoqing and EO-1 Hyperion datasets. As described in Table III, FastHyMix and FGLR, respectively, achieve the fastest and second fastest running time. However, this computational advantage comes at the expense of restoration quality, as FastHyMix and FGLR exhibit limited ability in suppressing severe mixed noise. Although LLRSGSSTM shows a relatively higher computational overhead, it delivers vastly superior denoising performance when dealing with heavy hybrid noise degradation, which demonstrates that LLRSGSSTM reaches a highly favorable balance between robust denoising efficacy and computational efficiency.

H. Computational Complexity Analysis

Finally, the computational complexity of LLRSGSSTM summarized in Algorithm 1 is analyzed in detail.

1) The complexity of $\mathcal{S}^{(k+1)}$ via (23) at each iteration is $O(n_1n_2n_3)$.
2) The complexity of $\mathcal{A}^{(k+1)}$ via (25) at each iteration is $O(n_1n_2rd^2)$.
3) The complexity of $\mathcal{U}^{(k+1)}$ via (27) at each iteration is $O(n_1n_2r)$.
4) The complexity of $\mathbf{V}^{(k+1)}$ via (29) at each iteration is $O(n_3r^2)$.
5) The complexity of $\mathcal{G}_1^{(k+1)}$ via (31) at each iteration is $O(n_1n_2n_3)$.
6) The complexity of $\mathcal{G}_i^{(k+1)}$ $(i = 2, 3)$ via (33) at each iteration is $O(n_1n_2n_3(\log(n_1n_2n_3) + \min(n_1, n_2)))$.
7) The complexity of $\mathcal{B}^{(k+1)}$ via (36) at each iteration is $O(n_1n_2n_3(\log(n_1n_2n_3)))$.
8) The complexity of $\mathcal{X}^{(k+1)}$ via (38) at each iteration is $O(n_1n_2n_3(\log(n_1n_2n_3)))$.
9) The complexity of $\mathcal{D}^{(k+1)}$ and $\mathcal{H}_i^{(k+1)}$ $(i = 1, 2, 3)$ via (39) at each iteration is $O(n_1n_2n_3)$, the complexity of $\mathcal{P}^{(k+1)}$ via (39) at each iteration is $O(n_1n_2r)$.

Therefore, the overall computational complexity of LLRSGSSTM in Algorithm 1 per iteration is $O(n_1n_2n_3(\log(n_1n_2n_3)+\min(n_1, n_2))+n_1n_2rd^2+n_1n_2n_3(\log(n_1n_2n_3))+n_1n_2n_3+n_1n_2r+n_3r^2)$.

V. CONCLUSION

Based on tensor subspace modeling, we propose a novel LLRSGSSTM method for guaranteed HSI destriping and denoising. On one hand, by leveraging the inherent directionality of stripe noise, we aim to explore the physical property of the multimode gradients of stripe noise, and propose the zero tensor property of mode-1 gradient tensor of stripe noise as well as the low-tubal-rank properties of mode-2 and mode-3 gradient tensors of stripe noise. On the other hand, by leveraging the spectral LR attribute of spectral gradient of HSI, we aim at learning the LR SGS of HSI with structure tensor model, which decomposes the spectral gradient of HSI into the SGSRC and the learnable SGSOBM. Moreover, the SGSRC indeed contains such distinct yet important spatial structures as corners, edges, and smooth regions, we novelly leverage the structure tensor to model these desired spatial structures of SGSRC and propose the unified SGSRC-based STV prior of HSI. Preferably, the SGSRC with much smaller size can lead to much lower computational complexity than the HSI or the spectral gradient of HSI. Subsequently, we leverage the ADMM algorithm to optimize the LLRSGSSTM model and theoretically prove the convergence of LLRSGSSTM algorithm. Finally, extensive experiments are shown to verify the superiority of LLRSGSSTM for HSI destriping and denoising.

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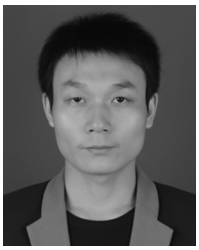
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